An End-to-End Machine Learning Framework for Credit Card Fraud Risk Management in Banking Systems

By

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Declaration

I hereby certify that this report constitutes my own work, that where the language of others is used, quotation marks so indicate, and that appropriate credit is given where I have used the language, ideas, expressions, or writings of others.

I declare that this report describes the original work that has not been previously presented for the award of any other degree of any other institution.

Name: Shavana Shaik

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Acknowledgement

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Abstract

The rapid growth of digital transactions and the increasing sophistication with which fraudulent activities are being carried out has made credit card fraud a significant challenge in modern financial systems. The financial datasets involved with fraudulent transactions are highly imbalanced, and this further complicates the detection process for these fraudulent cases which constitute only a tiny portion of transactions. This work provides an end-to-end analysis of credit card fraud detection using a combination of supervised learning, anomaly detection and explainable machine learning for full AI approach.

In the dataset used for this study, we have an anonymized credit card transactions and since features are transformed using Principal component analysis (PCA) so as to maintain privacy of data. We followed a standard data science pipeline structure including preprocessing, exploratory data analysis, model development and model evaluation. (4) Various machine learning models were applied: Logistic Regression were utilized as baseline, Random Forest and XGBoost for ensemble learning methods, and Isolation Forest (anomaly detection).

In order to deal with class imbalance, several methods were used: those included Synthetic Minority Over-sampling Technique (SMOTE) and cost sensitive learning. Additionally, combining anomaly scores from Isolation Forest to XGBoost yielded a hybrid model with improved performance in detecting rare fraudulent transactions.

Specifically, ensemble based methods flag a significant enhancement over the baseline model performed randomly while random forest does have an impressive precision and F1-score but also brings larger coverage while XGBOOST gives a numeric reasonable score between precision and recall. The hybrid model achieves additional robustness since it is based on supervised and unsupervised approaches. SHAP (SHapley Additive exPlanations) was also employed to interpret model predictions and gain insights regarding feature importance for enhancing the transparency of the models.

In conclusion, the proposed approach shows that a combination of multiple machine learning techniques along with an easy preprocessing step and effective interpretability methods provide a robust and efficient fraud detection system. The results emphasise the need for a trade-off between detection sensitivity and false positive rates, rendering the proposed model apt for use in real-world financial settings.

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Chapter 1: Introduction

1.1 Problem Description, Context and Motivation

In the modern financial ecosystem, credit card fraud has become an essential problem, as a result of the swift increase in digital transactions and e-payment systems. Fraud leads to massive economic losses to the financial institutions and consumers alike as it grows in complexity and scale. Naive approach such as rule-based system is no longer useful because they can not cope with novel and complex behavior of fraud. This situation increases the demand for supporting smart and data-driven approaches to detect fraudulent transactions [1], [4].

The most significant difficulty in credit card fraud detection is the high class imbalance in transaction datasets, where fraudulent transactions constitute a very small portion of overall data. This causes practitioner error with machine learning models to accurately learned negative and positive classes without being biased toward positive class. In addition, fraudulent behaviour is not always clear-cut, and appears to be indistinguishable from normal behaviour buried in high-dimensional data (for instance, when features are transformed using methods such as Principal Component Analysis (PCA) for preserving privacy [3]).

This project was motivated by the requirement to build an efficient fraud detection system that captures fraudulent transactions accurately without generating too many false positives. This means that a very high false positive rate can frustrate valid users while missed fraud cases can cause financial losses. Achieving a balance between detection accuracy and reliability is imperative for this reason. These challenges can be overcome through the use of machine learning, anomaly detection, and explainable artificial intelligence in a project aimed at enhancing fraud detection performance [1], [11].

1.2 Objectives

This project is therefore focused on development and implementation of an efficient credit card fraud detection system using trendy machine learning approaches. Through analysis within the dataset, they are expected to correctly classify the transactions as legitimate or fraudulent.

A main goal is to overcome the class imbalance problem by using effective techniques, like Synthetic Minority Oversampling Technique (SMOTE) and cost-sensitive learning that improves model performance on imbalanced datasets [8], [9]. Additionally, the project is focused on assessing and comparing several machine learning models (Logistic Regression Random Forest XGBoost) [5], [6].

Moreover, the project implements an anomaly detection method based on Isolation Forest to flag abnormal transaction behaviours. Anomaly-based detection is then combined with supervised training into a hybrid model which further enhances overall detection performance. In addition, this project includes the application of explainable AI methods such as SHAP to deliver transparency and interpretability of model predictions which is crucial for models used in real-world financial scenarios [12].

1.3 Methodology

The methodology used in this project is based on a data science pipeline, starting with collecting and understanding the data. The data used in this project are anonymized bank card transactions, where sensitive information has been masked using PCA transformation and the labels of the resulting features are from V1 to V28 [3].

The preprocessing part includes steps such as data cleaning — in particular checking for any missing values or duplicates, and scaling of the features (in our case to normalize the transaction amount). EDA is performed to see how data distributes, if there are visible patterns and relationships among the variables.

After data exploration, several machine learning models are applied. First a baseline model with Logistic Regression, and advanced ensemble methods including Random Forest and XGBoost is then developed, which have proven strong predictive performance on classification tasks [5], [6]. A second machine learning model used is based on Isolation Forest, which identifies anomalies in the dataset to pinpoint unusual, rare behaviours related to transactions [10].

To handle class imbalance, Synthetic Minority Over Sampling Technique (SMOTE) is performed on training data to create synthetic samples of the minority class to facilitate learning by the model [8]. Moreover, cost-sensitive learning techniques are applied to assign more weightage to the fraudulent transactions. A hybrid approach is implemented combining the outputs of anomaly scores produced by Isolation Forest with the predictions of the XGBoost model for better detection accuracy.

The ability of the classifier is validated by using metrics like precision, recall, F1-score and ROC-AUC to evaluate its classification performance. In the end, model prediction interpretation is performed through SHAP to distinguish essential features contributing to fraud detection [12].

1.4 Legal, Social, Ethical and Professional Issues

Machine learning models used for credit card fraud detection need to take a number of legal, ethical, social and professional issues into account due the sensitivity surrounding financial transaction data. Although the data set used in this project has been anonymised, it contains actual financial behaviour that must be treated appropriately while adhering to data protection principles.

At a legal level, financial data is vulnerable by its very nature and must adhere to severe restrictions (the likes of GDPR). Although the dataset used in this study do not contain any personal identifiable information, principles of data protection must be followed so that no attempt is made to re-identify someone or to use the data for research other than its intended purpose. The principles of data minimisation and purpose limitation must apply, namely: Only those features relevant to the specific context in which fraud detection is taking place should be used, whilst the data must not be misused outside of that context.

When it comes to the ethics of fraud detection systems, the bias along with unfair decision making is one of the main concerns. If a machine learning model is built in an imbalanced manner, it can become biased and be unable to classify properly on the less representative class. In this respect, false positives- genuine transactions flagged as fraud, hinder customers through blown-out inconveniences, haltered transactions/money and eroded trust on financial institutions. On the other side, a false negative can cost money because a fraud goes unnoticed. Hence, a tradeoff between precision and recall needs to be observed in order to have just value results.

A third ethical aspect is transparency and explainability. Decision justification is crucial for financial systems, particularly when being applied to deny transactions or flag suspicious activity. SHAP-based explainability -- Interpreting model predictions and finding key features affecting decisions This builds confidence and accountability and conforms with responsible artificial intelligence.

On the other side, Fraud detection systems also play an important role in society as they help protect people and organizations from financial crime. Well-performing models bring security and trust to digital payment frameworks. However, if machines start laying down our rules and we just follow along, then that may have unintended consequences and to avoid this we need to balance machine decision-making with human judgment or mindset.

On the professional side, this project follows best practices about data science (proper handling of data, model validation and performance tracking). It is that the business of all sources of considerable value and are to be cited accurately Academic integrity, and even though results are

honest Reporting methods, without manipulation. Furthermore, the implementation is reproducible and can be tracked and verified for its models/core results.

Third, legal compliance, fairness (when ethical), traceability and ethics when required are also approaches that should be taken mind of us maintaining the responsible development of frames for fraud detection. All these factors need to be addressed to ensure that the solution being proposed is both technically sound as well as socially responsible and reliable.

1.5 Background

The issue of credit card fraud detection has been widely addressed in recent literature because of the practical significance and difficulty of working with imbalanced data sets. Early methods used statistical and rule-based systems, which could not adapt to new types of fraudulent behavior. These advancements in machine learning have resulted in more complex models aimed at improving detection performance [4], [16].

This dataset has undergone PCA transformation using your transactions, and the column features will give you anonymous labels of V1 to V28. By transforming the data in this, we largely eliminate concerns of being able to identify any user or their actions while keeping essential information about how users are using it. Also, more features like time and amount of each transaction help in their analysis.

Ensemble methods like Random Forest and XGBoost have been shown [5], [6] to be effective for different fraud detection tasks. Anomaly detection methods, for example, have also been utilized to sniff rare and abnormal patterns in data [10]. Explainable AI — particularly SHAP: approaches have also gained traction for providing transparency, by enhancing model interpretability [12].

In this project, we further improve upon these existing approaches by integrating multiple models and additional hybrid framework with the goal of mutual improvement on performance as well as interpretability.

1.6 Structure of Report

This report is organized into five main chapters. Chapter 1 provides an overview of the problem, objectives, methodology, and key considerations of the project. Chapter 2 presents a comprehensive literature review and discusses the technologies used in fraud detection.

Chapter 3 describes the implementation of the system, including data preprocessing, exploratory data analysis, and the development of machine learning models. Chapter 4 presents the evaluation and results, including model performance comparisons and analysis. Finally, Chapter 5 concludes the report by summarizing key findings, discussing limitations, and suggesting directions for future work.

2. Literature and Technology Review

2.1 Literature Review

The rising number of digital transactions and the sophistication of fraudulent behavior is leading to credit card fraud detection being a major area of study. Fraud detection in traditional systems were largely rule-based and statistical methods, which have limitations in adapting to new and evolving fraud patterns. With the development of machine learning techniques, more intelligent and adaptive systems have been developed to enhance detection accuracy and efficiency [1], [4].

The imbalanced class distribution is a big challenge of the fraud detection problem as fraudulent transactions are rare in the dataset. Such a distribution can materially impact the performance of machine learning models, which can be biased toward the majority class. Dal Pozzolo et al. [3] stressed the need of realistic modelling methods for fraud detection noting the difficulties in highly imbalanced datasets. In order to solve this problem, several techniques have been proposed such as oversampling, undersampling and cost-sensitive learning.

One of these techniques, which has been proposed by Chawla et al. [8], is among the most common approaches. SMOTE works by creating synthetic instances of the minority class, thus balancing the data and enhancing model capability. Learning from imbalanced data was also further studied by him and Garcia [9], studying methods to improve classification performance for the different learning practices.

Machine-learning models have been widely used across fraud detection systems. A baseline model which is used and simple to interpret: Logistic Regression However, in the case of higher dimensional data with a more complex form of relationship it might fail. Ensemble learning approaches like Random Forest were found to work well as they build an ensemble of decision trees increasing the predictive power and mitigating overfitting [5]. Another well-known boosting algorithm is XGBoost, which has become widely-used due to its efficient, scalable approach and high performance on classification problems [6].

Alongside supervised learning approaches, anomaly detection methods were frequently applied in parallel to detecting fraudulent transactions. Fraudulent behaviour is also rare and differs significantly from normal patterns, making it a likely candidate for anomaly detection. Isolation Forest Liu et al. [10], which is one of the popular anomaly detection algorithm that isolates anomalies based on their uniqueness. Chandola et al. Another finding [11] was an overview of anomaly detection and showed how normal patterns can be found in big datasets;

Several recent studies have proposed hybrid methods that use more than one technique to enhance fraud detection performance. These methods utilize various algorithms complementary to one another for improved accuracy and robustness. Mienye et al. Another approach proposed in [14] is combining the outputs of several classifiers (ensemble and hybrid models) that showed a significant improvement in fraud detection. Likewise, West and Bhattacharya [15] stressed the unified use of various machine learning methods to combat dynamic financial scams.

One more critical advancement in fraud detection research is the combination of explainable artificial intelligence (XAI) with machine learning. As ML models grow increasingly complex, interpretability of model predictions becomes an important requirement in applications such as finance where transparency and accountability are needed. SHAP (SHapley Additive exPlanations) was introduced by Lundberg and Lee [12] as a method that allows inference of the contribution on model predictions attributed to each feature. SHAP is a consistent method to understand complex models like ensemble methods and has been widely adopted.

In conclusion, the findings of studies preclude a single technique to solve all fraud detection problems. Instead, we need to build a system that combines multiple machine learning models, anomaly detection algorithms and explanation methods in order to be effective and reliable. This work expands on existing research into these avenues by combining multiple models and creating a hybrid framework to enhance fraud detection performance.

2.2 Technology Review

Various technologies and tools used in data processing, machine learning and visualization constitute the implementation of this project. For example, Python was chosen as the main language because it has a rich set of libraries and strong support for applications in data science. It lets you work with big datasets and algorithms in a more efficient way.Analyzed Using KnnP 288

The code for this project is based on Google Colab, a cloud-based Python development environment. As Email to translate in natural language: Google Colab unlocks the power of large-

scale data analysis and model training, but without needing local installation. The platform also supports google drive integration where you can easily access datasets and project files available in the cloud.

For the entire project, several Python libraries were involved to achieve maximum performance. It make us use Pandas and NumPy for data manipulation and numerical computation. These libraries contain optimized data structures and methods to work with large datasets. It is also a higher-level interface to know for the underlying Matplotlib and Seaborn packages, allowing us to generate informative plots with commendable aesthetics during exploratory data analysis.

Scikit-learn methods were employed to construct and assess models like Logistic Regression, Random Forest for executing machine learning. Reinforcement Learning methods while finding their place in the field of artificial intelligence. Again, for advanced modelling XGBoost [6] a powerful gradient boosting library, which provides high performance and scalability was used. SMOTE [8] was applied using the imbalaced-learn library to solve the class imbalance within the dataset.

The anomaly detection for identifying unusual transaction patterns was performed using Isolation Forest from Scikit-learn [10]. Additionally, SHAP (SHapley Additive exPlanations) to interpret model predictions and gain insights into feature importance [12]. SHAP increases model transparency and is especially useful in financial use cases where interpretability is paramount.

By leveraging these technologies, we can build a powerful platform for developing and testing machine learning approaches to fraud detection. This reinforces the project in terms of reliability, scalability and reproducibility by showing that the tools chosen are commonly used across industry and research.

2.3 Summary

This chapter presented the summary of existing research and technologies on credit card fraud detection. These existing works emphasize the issues of class imbalance and indicate that improved detection performance can be accomplished by leveraging sophisticated machine learning methods. To download the research paper on handling class imbalance in image datasets using SMOTE, ensemble learning, anomaly detection and explainable AI, please visit:

In addition, the technology review described the tools and framework used to complete this project such as Python and Google Colab then various machine learning libraries. For data processing, model development and evaluation these technologies provide the way forward.

From the literature and technology studying analysis, it can be seen that integrating multiple approaches can be a better solution for fraud detection systems. This work takes such an approach, employing supervised learning, anomaly detection and explainability methods to form a complete and thorough fraud detection model.

Chapter 3: Implementation

3.1 Data Preprocessing

This step is critical in creating an efficient fraud detection system, since the quality and structure of the data directly affect how well a model will perform. The data in this project is credit card transactions, but all sensitive information has been converted into principal components using PCA. This leads to features named from V1 to V28, which are the main components of the original data while keeping privacy [3].

The first thing was loading the dataset into the Google Colab environment and inspecting it to understand its format/explore. After that, some basic data exploring techniques were implemented: the number of rows and columns, the type of data it had in each column or if there were null values. Moreover from the analysis we confirmed that there were no missing values in our dataset guaranteeing a total completeness of data. To prevent redundancy and maintain data integrity, duplicate records were additionally cross-verified and eliminated.

Feature scaling was one of the important preprocessing steps. The transaction amount feature had a large range and was highly multimodal. This was resolved by standardization (using scaling methods to normalize the distribution) This means that the feature will impact model training proportionally and not dominate other features.

Feature selection was another key mean of preprocessing that we employed. The dataset had a time-related feature, which in this case was irrelevant to detect a fraud. Hence it was removed to reduce the dataset complexity and Increase Model Efficiency. The models learn better patterns because they concentrate on the most related features.

One of the major challenges in this dataset, is extreme class imbalance - fraudulent transactions account for only a small fraction of the data. This can result in imbalanced models that only predict the majority class. To handle this problem, we applied the Synthetic Minority Oversampling Technique (SMOTE) to the training data. SMOTE is used to generate synthetic samples of the minority class, which balances the data set and improves model performance [8].

Advanced models like XGBoost also integrated cost-sensitive learning by assigning greater weight to fraudulent transactions in addition to using SMOTE. Basically, we treat the rare class with a little more consideration during modelling and thus increase our predictive ability to identify this type of fraud.

The data set was then divided into training and testing sets applying stratified sampling to preserve the original class distribution. This not only makes sure that both the sets represent possible values of the dataset but also enables an accurate model evaluation.

In summary, the preprocessing step prepared a clean and balanced dataset for machine learning. This provided a groundwork to build accurate and reliable models for fraud detection by tackling issues of feature scaling, class imbalance and data quality.

3.2 Exploratory Data Analysis (EDA)

3.2.1 Data Analysis on Raw Data (Statistical Analysis)

Before using any machine learning models in the dataset, we performed statistical analysis to examine the distribution of the dataset itself, for example — how it is distributed over certain distributions and what are its characteristics that make imbalance on the data. It contains 284,807 transactions and 31 features (anonymised variables V1–V28, Time, Amount and the target Class).

The summary statistics show that almost all anonymised features have a mean ~0 and a unit variance, which reflects our expectation since they were created using Principal Component Analysis. The next two columns — Time (ranging from 0 — 172,792 seconds as a temporal sequence) and Amount (an intermediate feature with high variance and wide range, ranging from $0–$25,691.16 with mean around $88.35 and median of $22.00). The fact that the average is so high, far higher than the median, indicates this distribution is very skewed.

This is further validated by the skewness value of the Amount Feature of ≈ 16.98 and its kurtosis value of >845. This is representative of a heavy-tailed distribution with extreme outliers, i.e. high number of transactions in lower range and a small number of very high value amounts

The extreme imbalance of the classes is a fatal attribute of the dataset. Of 284,807 transaction all are not fraud transactions only 492 it is which account for about 0.17 per cent of the dataset while they were the rest of legitimate (99.83) transactions. The imbalance in this case could pose a problem to any machine learning algorithms, since they may be oblivious and tuned towards most of the data (majority class), which means fraud cases are difficult to detect.

Statistical comparison between fraudulent and legit transactions shows different characteristics. The average transaction amount for fraudulent transactions is 122.21, whereas it is 88.29 for genuine transactions. Interestingly, features like V1, V2, V3 and V4 show a large difference in mean values between class 0 and class 1 suggesting their role in distinguishing between the two classes right away.

We corroborated these with correlation analysis as well to find the features that are more highly correlated with the target variable. In final data set, V14, V17, V12 and V10 have comparatively higher correlation with Class varaible suggesting their importance in fraud detection. Unlike Amount and Time, their correlation with fraud is very low meaning in isolation they are weak predictors of being fraudulent.

Table: Transaction Amount Statistics

Statistic	Value
Mean	88.35
Median	22.00
Standard Deviation	250.12
Minimum	0.00
Maximum	25,691.16
Skewness	16.98
Kurtosis	845.08

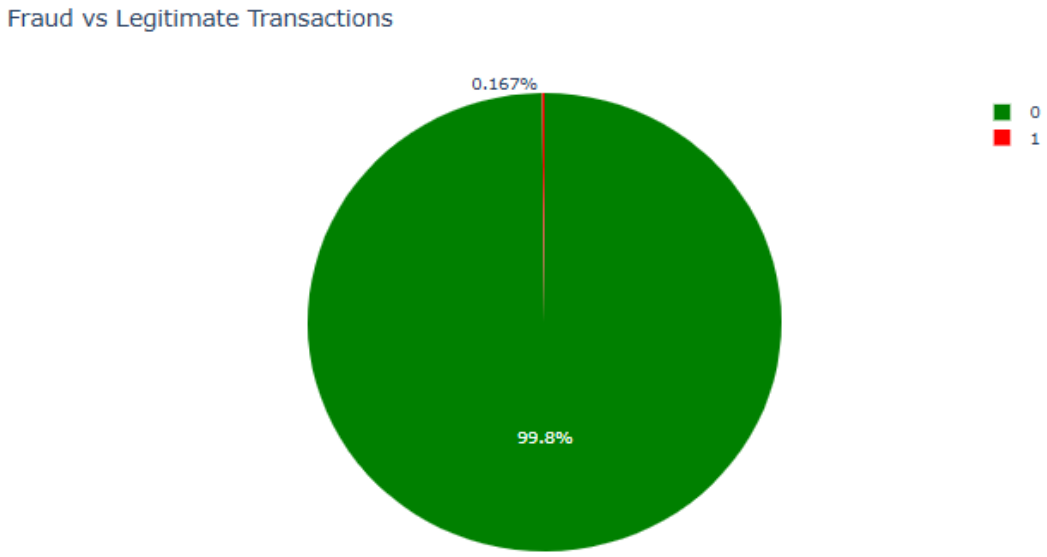
The data was analyzed statistically and leads to three general observations. The first is that the dataset is severely imbalanced, so SMOTE is to be used for better learning of models. Secondly, the transaction amount distribution is incredibly imbalanced with massive outliers so scaling is needed for it to be modelled effectively. Third, some anonymized features are highly discriminatory, and these immediately come into play in flagging the transaction as fraudulent.

These insights lay the groundwork for modelling phases that follow and are a basis to understand the preprocessing and modelling decisions made in this project.

This step is very important for understanding the structure, distribution and relations inside the data before trying machine learning models on it. It assists in recognizing trends, abnormalities, and subsets responsible for fraud observation. Various visualizations were used in this project to analyze the dataset and draw useful insights.

The analysis began with the distribution of target variable.

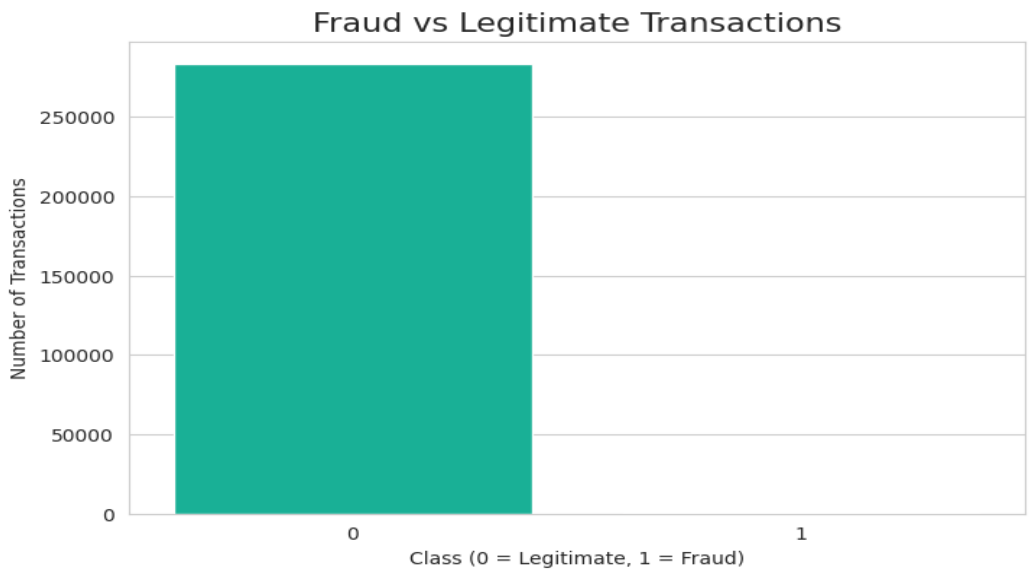
Figure 1: Fraudulent v/s Legitimate Transactions (pie chart)



The pie chart illustrates the distribution of classes in the dataset, indicating that it is heavily unbalanced: around 99.8% of transactions are labelled legitimate and only a fraction represents fraudulent cases. This discrepancy presents a challenging scenario for classification models, leading to the need for techniques such as SMOTE to enhance detection effectiveness [8].

To further verify this observation, the class distribution was analyzed in terms of a bar chart visualization.

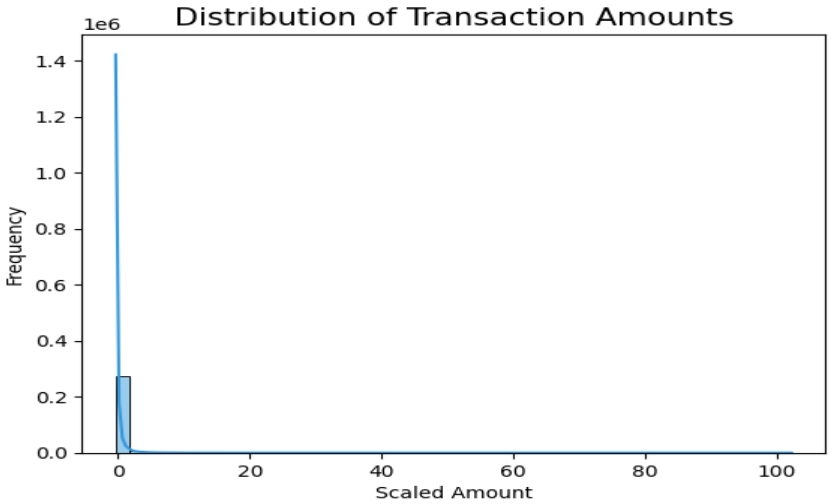
Figure 2: Number of transactions by class



The bar chart validates the prevalence of legitimate transactions over fraudulent ones. This imbalance indicates that simply achieving accuracy may not be an appropriate evaluation approach, rather this considers using measures such as recall and F1-score to evaluate model performance.

It also analyzed the distribution of transaction amounts, so it could see how people were spending.

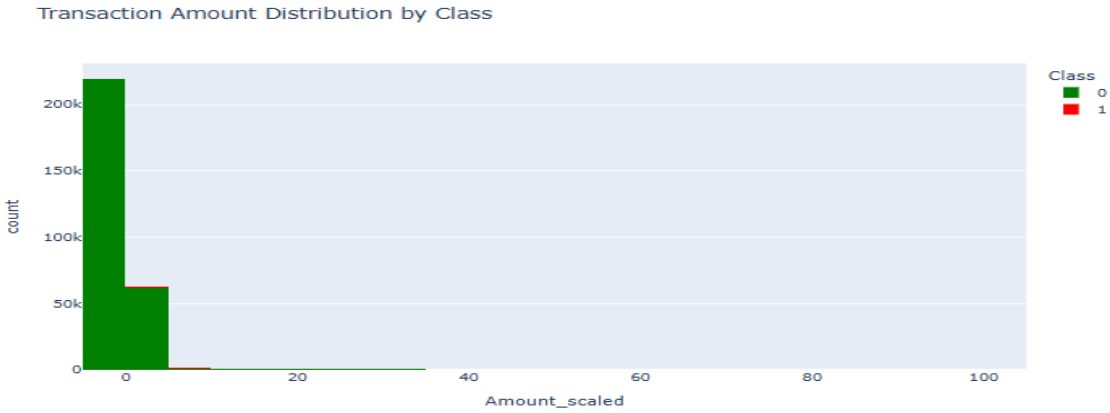
Figure 3: Scaled transaction amounts distribution



The histogram demonstrates a highly skewed distribution for most transactions at lower values with few transactions occurring have high values. This suggests that we have outliers and validates the use of scaling techniques in preprocessing.

And to visualize a difference in the transaction amounts between fraud and legit transactions, a class-wise distribution was plotted.

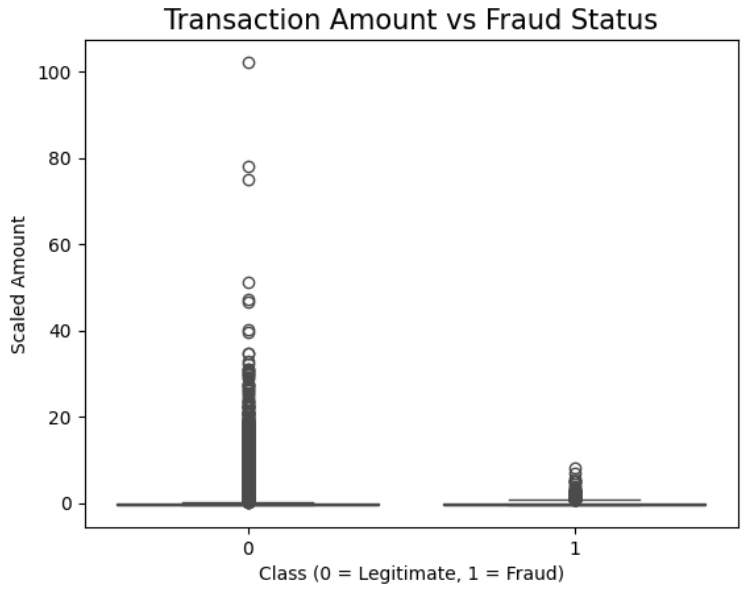
Figure 4: A histogram of transaction amounts per class



Both fraudulent and legitimate transactions appear to be concentrated in lower value ranges, but the exact distribution differs slightly between these two types of transactions. Clearly, this leaves room to use only transaction amount as a fraud differentiator.

So a scatter plot was also made, in order to analyze how the transaction amount correlates with fraud.

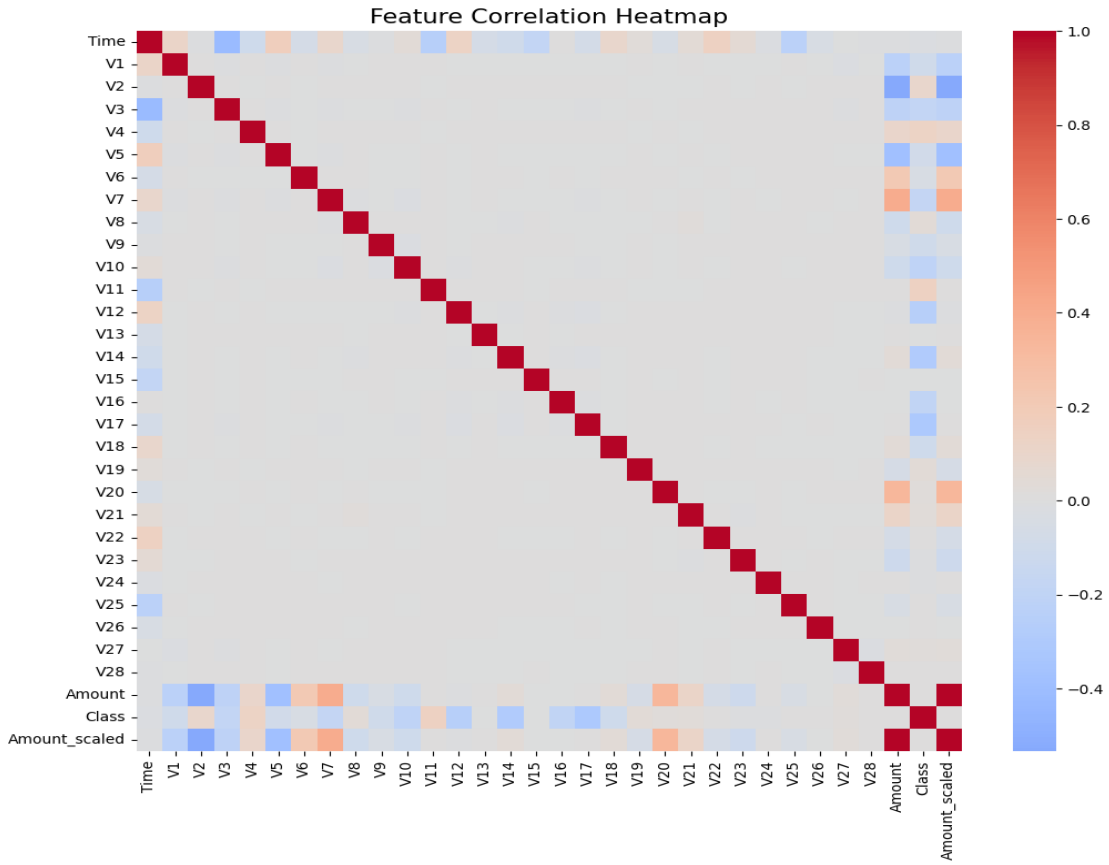
Figure 5: Transaction amount and fraud status



The separation cloud plot shows that the fraudulent transactions are spread across differing transaction amounts with no apparent clear separation. This highlights the necessity for sophisticated machine learning models that can discern such intricate complexities.

A correlation heatmap was produced to better understand the relationships between features.

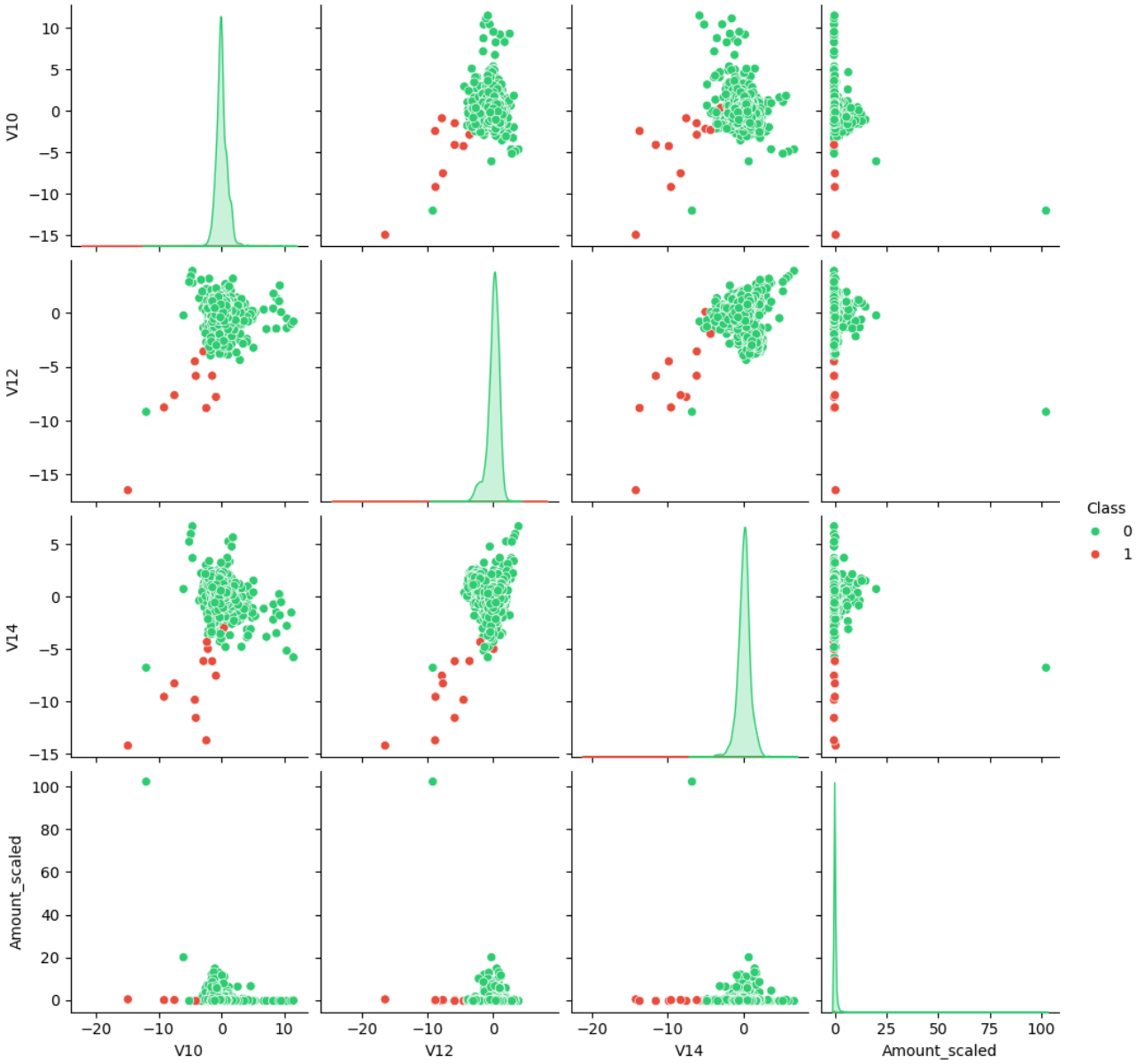
Figure 6: Correlation matrix of features



The heatmap indicates that a vast majority of features do not correlate which is expected as dataset has been reduced by PCA transformation [3]. However, some features correlate with the target variable suggesting they may be of importance on fraud detection.

Individual features were further analyzed to see their behavior in fraudulent transactions.

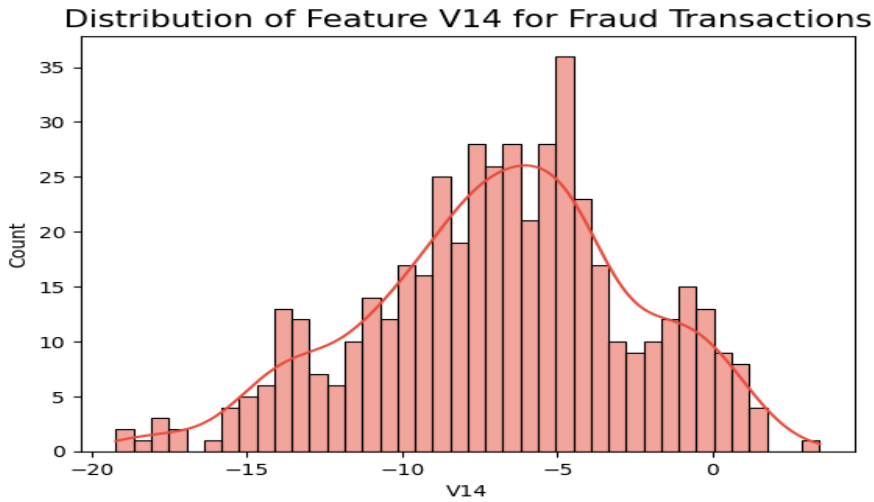
Figure 7: Pairwise relationships between features of interest.



The pair plot depicts the clustering pattern, where we can identify clustered groups for legitimate transactions and scattered outliers for fraudulent transactions. This enables the application of anomaly detection methods, e.g., Isolation Forest [10].

Additional scatter plots were examined to understand feature behaviour further.

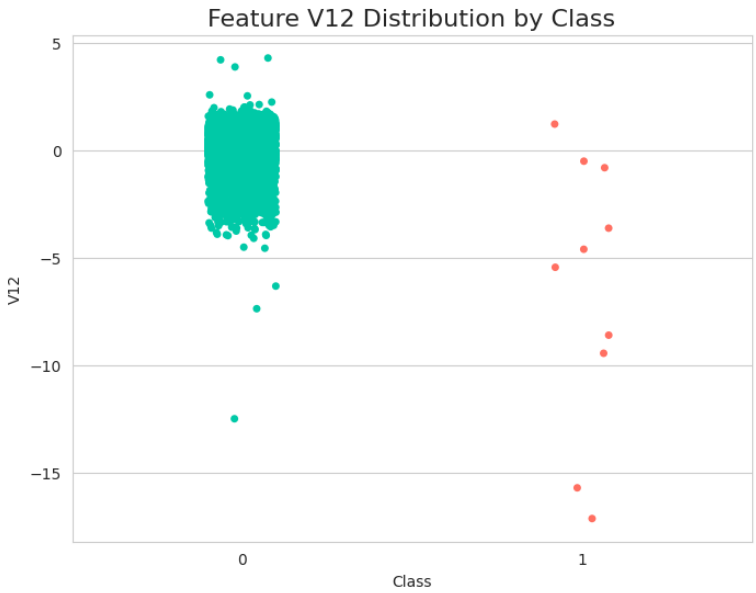
Figure 8: Distribution of feature V14 for fraudulent transactions



The histogram for feature V14 shows a unique pattern in the case of fraud, suggesting that this is one of the most impactful features that determines whether a transaction was fraudulent or not. This is later verified by the model-based feature importance analysis.

A pairplot was used to examine relationships among multiple features at once.

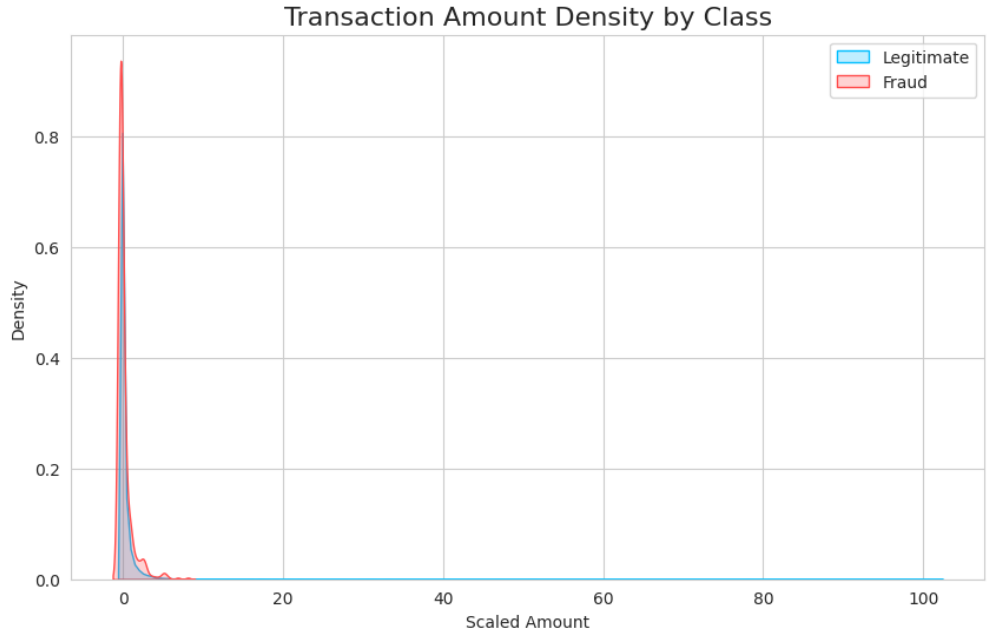
Figure 9: Feature V12 distribution across all transaction classes



The fitted line is "far" from the normal points which refers to fraudulent transaction that have many V12 values contrarians showing why this predictor is also relevant for classification.

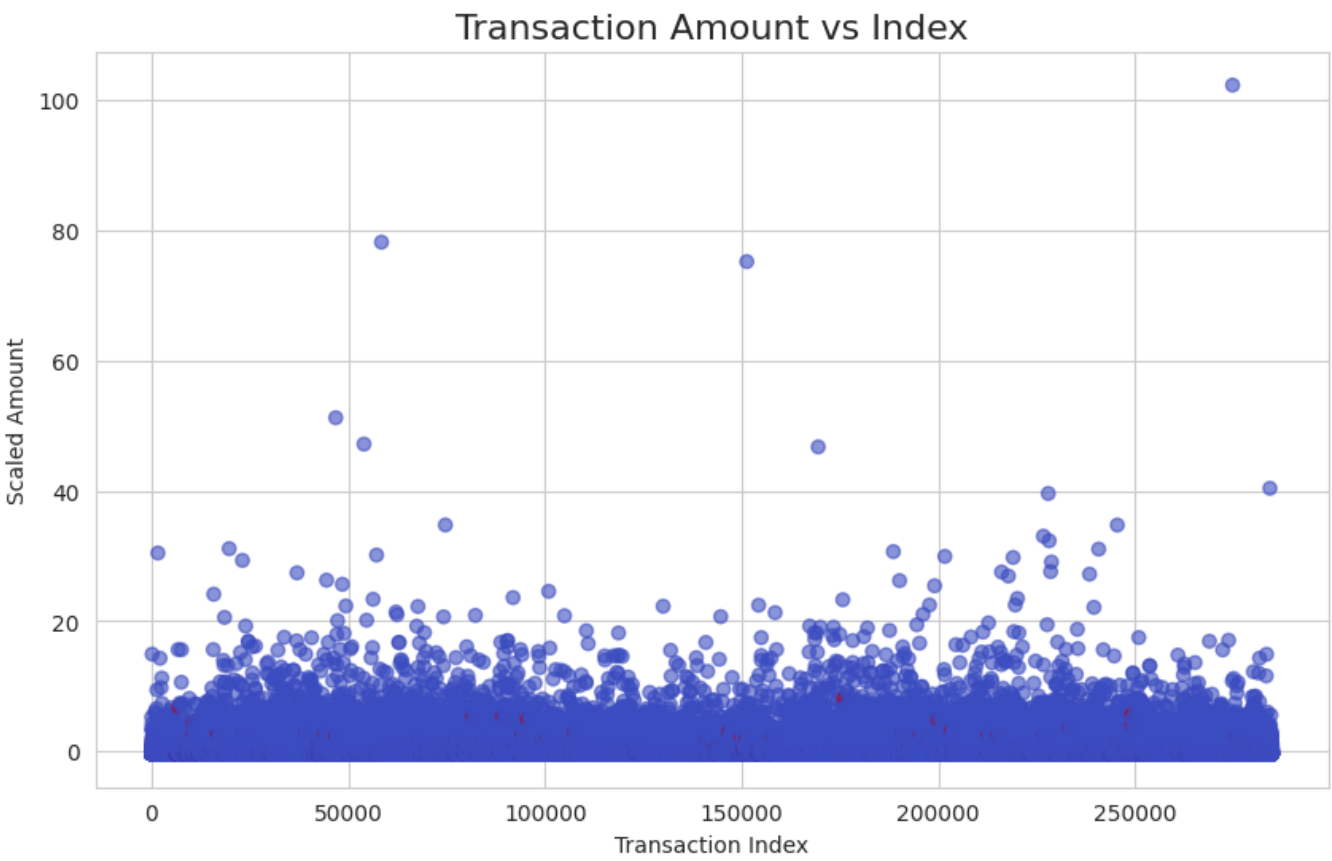
Lastly, a density plot was applied to compare transaction distributions across classes.

Figure 10 : The density distribution of transaction amounts with the class



The density plot above shows marked overlap indicating that there is no strong separation between fraudulent and legitimate transactions. The overlap also demonstrates a key point: fraud detection is not a simple threshold-based problem.

Figure 3.11: Scatter plot showing the distribution of scaled transaction amounts across transaction index.



The scatter plot indicates that during an initial period of time, most transactions are concentrated on the lower end of the scale and a few high-value outliers across the dataset. This is similar to the approach followed where we take a temporal index range over which no fraud occurs at all. This shows the importance of building models to identify irregularities instead of simple rules.

In conclusion, the exploratory data analysis showed a lot of interesting points. There is a large class imbalance, and many fraudulent transactions are indistinguishable from legitimate ones. Some of these features, like V14 and V12, show strong discrimination patterns so they're useful to train the model. Moreover, the existence of anomalies and outlier patterns make hybrid methods integrating supervised learning with anomaly detection an appropriate approach. This information informed the choice of preprocessing techniques and machine learning models used in later phases of the project.

3.3 Model Implementation

In the model implementation phase, we have to train multiple machine learning models on different prepared data. So you can see, in this project he made sure to clearly define the input features and target variable as it relates to your model training prep steps.

It contains anonymised features V1 to V28 as well as Time, Amount and the target variable Class. The model is already scaled and ready for modelling as the features V1 to V28 are created with Principal Component Analysis. The Amount feature, on the other hand, was standardised separately for all features to be fully consistent.

Time was removed from modelling, being non-predictive of fraudulent activity in this case. This makes it easier to remove the feature without being degraded in performance, as well eliminates complexity and increase computing efficiency.

The final input dataset for model training looks like this:

- Input features X comprising of V1 to V28 and scaled Amount
- Target variable y is the Class column, which takes the value 0 for legitimate transactions and the value 1 for fraudulent transactions

Once the input variables have been defined, stratified sampling was used to split the dataset into training and testing sets. This keeps the fraudulent vs legitimate transaction distributions the same across both sets. For example, they set aside 80 percent of the data to train the model and 20 percent for testing it.

Given the very high class imbalance in dataset, extra pre processing was applied to help model learn the dataset better. Synthetic Minority Oversampling Technique was used ONLY on the training data. These create synthetic examples of needle transactions that will help the models learn patterns better, but not adding any bias to the testing data. It should also be noted that we do not apply SMOTE to test set, because it will break the edge model evaluation process.

The models used in this study are Logistic Regression, Random Forest, XGBoost, Isolation Forest and hybrid model. To allow for a fairly apples to apples comparison of performance, each model receives the same set of K input features.

The baseline model was created using Logistic Regression which got the initial training set of preprocessed data. Using the same inputs, Random Forest and XGBoost were then fitted, utilizing ensemble and boosting techniques to better capture complex relationships in the data.

Since Isolation Forest is an unsupervised model, it was fit on the feature set without using any information about the target. It generates anomaly scores of every transaction at the end, indicating how normal a transaction is compared to others in the dataset.

These Anomaly scores were then used as a feature in the hybrid model. In particular, the anomaly score created fromIsolation Forest was added to the input feature set and fed into XGBoost. This enables the hybrid model to take advantage of both supervised learning and anomali detection.

The input configuration for the models helps ensure all of them are trained on consistent, well-prepared data. Improper data splitting, feature scaling & class imbalance handling provides a strong foundation for the accuracy and reliability of fraud detection.

The next phase involved implementation of the model that is at the heart of this project with a few different machine learning techniques used to predict fraudulent credit card transactions. Adhering to the avowed highly imbalanced nature of dataset, an ensemble model with baseline models was built along with adding isolation forest for anomalous predictions. Various models were chosen to cater to different patterns in the data including but not limited to linear relationships, non-linear behaviors and outliers.

Logistic Regression was first implemented as the simplest and probably the most interpretable model. While it performed well for the linear relationship it was able to capture, it struggled due to the hard imbalance in the data available. It underscored the necessity for more sophisticated approaches that could tackle imbalanced data distributions.

Ensemble learning methods are developed to overcome these limitations such as Random Forest. By combining many [6] decision trees to make predictions and reducing the variance, Random Forest improves the prediction accuracy in comparison to a single tree, proving its efficacy as a solution capable of managing complex datasets characterized by non-linear relationships. The performance of fraud detection was comparably improved over the baseline model.

This was further tuned with XGBoost, a scalable machine learning system for tree boosting that has consistently shown state-of-the-art results in various types of predictive modelling. XGBoost enhances model learning by incrementally constructing trees and incorporating regularization methods to effectively learn complex relationships in the data [7]. This model also performed robustly with respect to balancing precision and recall, indicating a high suitability for fraud detection tasks.

Beyond supervised learning methods, an unsupervised anomaly detection method, Isolation Forest, was deployed. This model detects anomalies by isolating observations that deviate_maximally from other observation and is called One-class SVM [10]. As most of the transactions are normal and fraudulent transactions are rare in nature with different behavior, Isolation Forest helped us a lot as it gave us meaningful results by giving an anomaly score to each transaction.

A hybrid model was created using Isolation Forest and XGBoost in order to increase detection performance. The outlier scores produced by Isolation Forest were utilized as an extra feature, which improved the model based on its ability to identify valid and fraudulent transactions. We achieve improved classification performance by combining an anomaly detection and supervised learning based approach.

To deal with class imbalance, the synthetic minority over-sampling technique (SMOTE) was used. Thus, SMOTE creates synthetic samples of the minority class that help these models learn better from fraudulent transactions [9]. Moreover cost-sensitive learning methods were applied to give higher penalty for the misclassification of fraudulent instances, allowing to obtain models which focus on fraud detection.

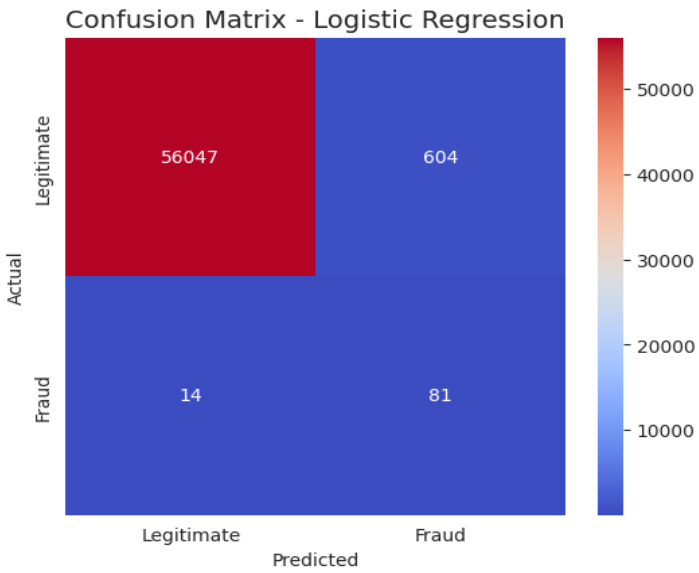
Model interpretability was ultimately tackled using SHAP (SHapley Additive exPlanations), which shows us the feature importance, and how each input affects the output. [11] SHAP values explain how contribution to predictions from individual features and thus give more transparency/trust on the edge.

In summary, this multi-model scheme and the use of sophisticated preprocessing techniques and interpretation methods provide a complete solution for fraud detection. Detailed Analysis for Each Model and Performance

3.3.1 Logistic Regression

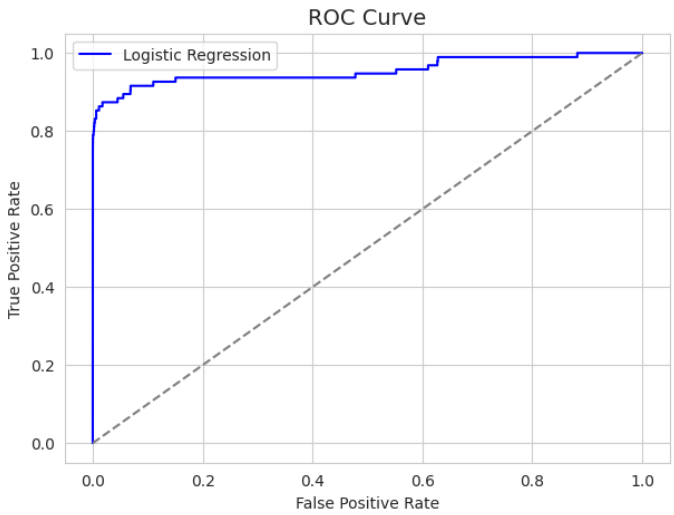
Initially, we implemented Logistic Regression as baseline to create an initial benchmark for fraud detection. Because of its simplicity and interpretability, it is commonly applied to binary classification problems especially when relationships among features are believed to be linear [5]. But, class imbalances significantly affect its performance, and that is a key issue with this dataset.

Figure 12: Confusion Matrix – Logistic Regression



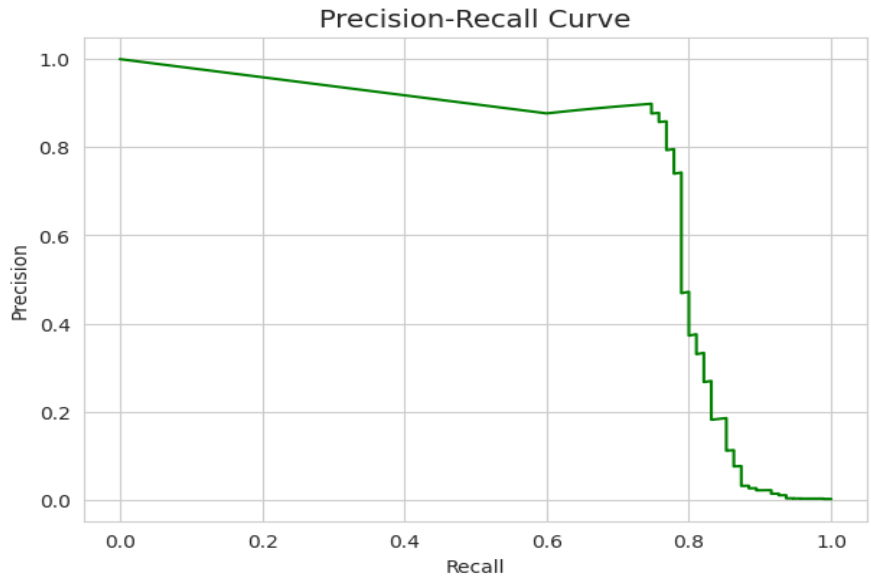
From the confusion matrix we can observe that the model correctly classified many legitimate transactions as well (True Negatives: 56,047). Still, it generated a significant number of false positives (604), or legitimate purchases being incorrectly classified as fraudulent. Notably, the model was able to detect 81 fraudulent transactions with just 14 misclassifications. This means that the model has a reasonably high recall while detecting fraud, but his precision suffers with false positives.

Figures 12: ROC Curve – Logistic Regression



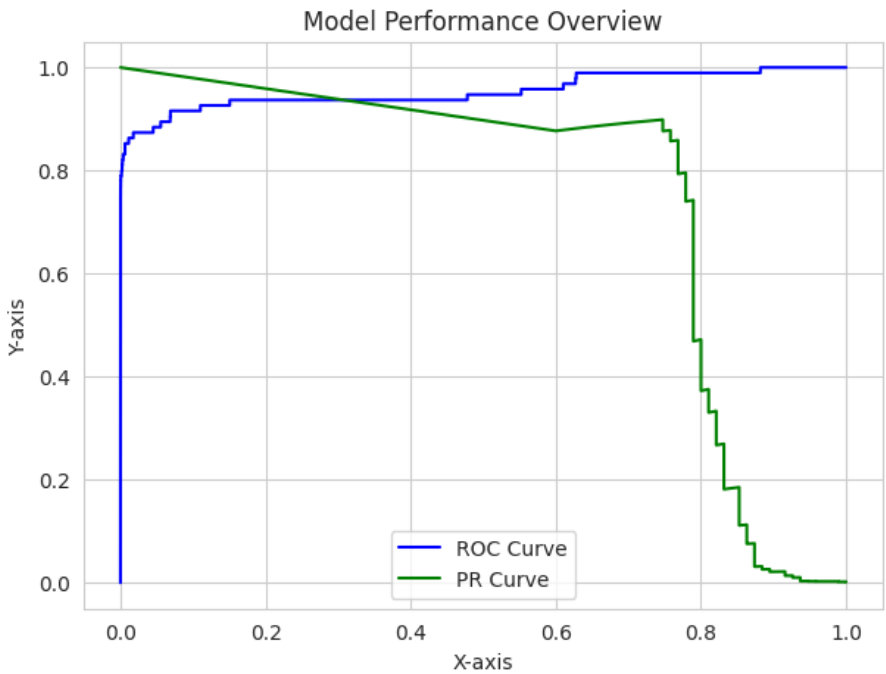
The ROC curve indicates performance, being pushed to the top-left corner. This means that the true positive rate is high and the false positive rate is low for a varity of thresholds. These numbers once again show that the model will be able to differentiate between a fraudulent transaction and a legitimate transaction.

Figure 14: Precision-Recall Curve – Logistic Regression



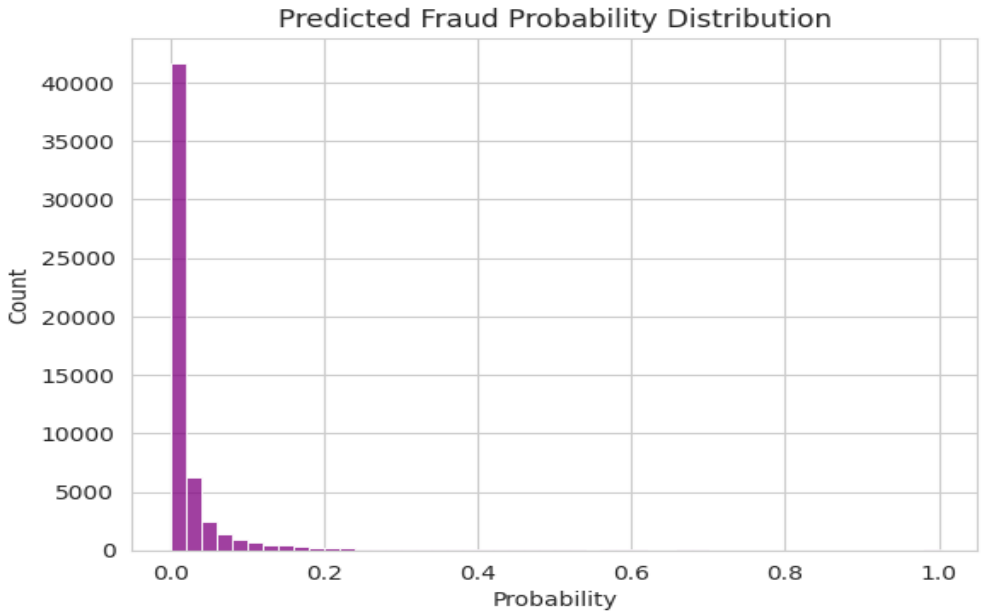
The tradeoff between precision and recall. The model has high precision at low recall, but performance slips as the recall goes up. Such occurrence is common in imbalanced datasets since high recall is usually accompanied by lower precision [8].

Figure 15: Combined ROC and PR Curve



And in this case the combined visualization gives us a better overview of our model performance. The ROC curve presents good classification behaviour, while the Precision-Recall curve shows it is difficult to identify rare cases of fraud. It demonstrates that, in the context of imbalanced datasets, ROC is not a enough criterion to evaluate performance.

Figure 16: Predicted Fraud Probability Distribution



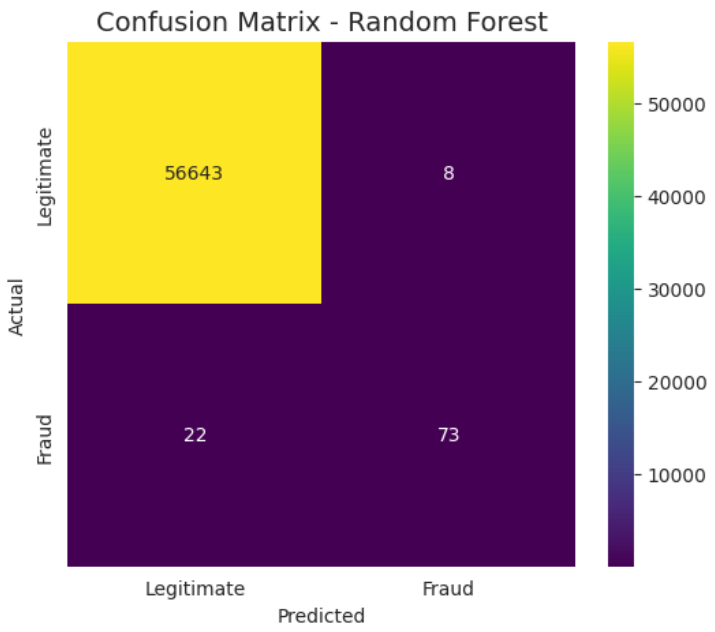
This means that the fraud probability distribution is such that most transactions get a very low score, while only few are getting higher scores. This is consistent with the data imbalance as it shows that the model is conservative in predicting fraud; however, this may also result in missed detections if thresholds are not properly tuned.

In general, Logistic Regression seems to be great baseline with decent recall for fraud detection. Its inability to handle class imbalance and non-linear relationships demonstrate the need to migrate towards better models such as Random Forest, and XGBoost discussed in later sections.

3.3.2 Random Forest

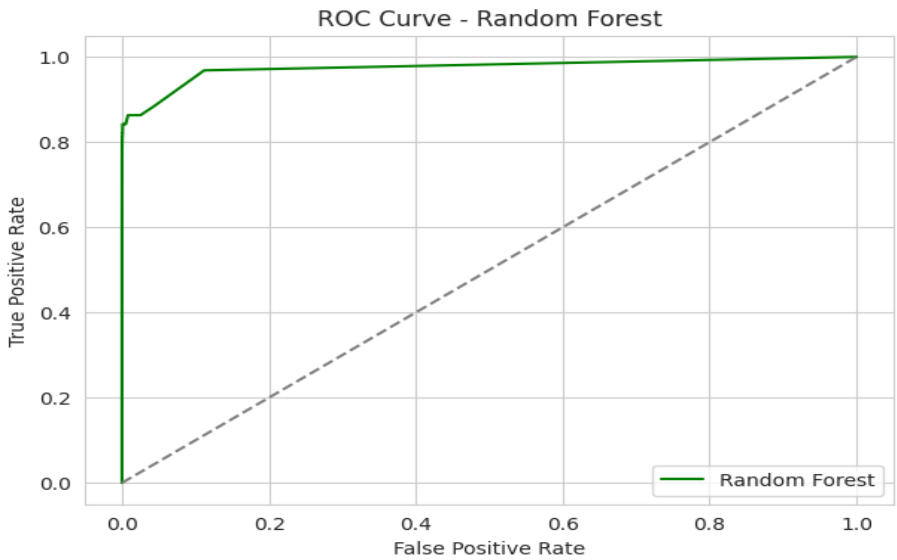
Random Forest is an improved ensemble learning algorithm which we implemented to improve the Fraud how detection performance of the base line Logistic Regression model. This algorithm works by training many decision trees and combining their predictions to generate a final output. Random Forest works by building a number of weak learners —and combines them together to reduce variance and enhance generalization [6]. Moreover, it is well equipped to manage non-linear relationships and interactions between features that are prevalent in fraud detection problems.

Figure 17: Random forest confusion matrix



The confusion matrix shows a significant improvement in classification performance when compared with Logistic Regression. An accuracy of 56,643 True Negatives were correctly identified with only 8 false positives suggesting a very high precision. This reduction in false alarms is especially important in actual financial systems, where legitimately flagged transactions can potentially annoy customers. Moreover, the model was able to identify 73 fraudulent transactions (True Positives) and misclassified only 22 fraudulent samples (False Negatives). This balance emphasizes the model's effectiveness in identifying fraudulent activity and minimizing false positives.

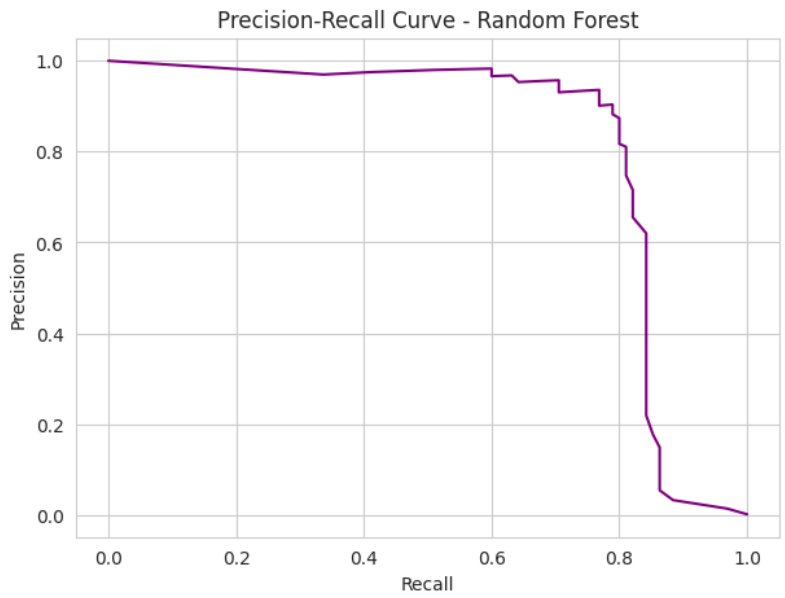
Figure 18 ROC Curve – Random Forest



The ROC curve demonstrates the good discriminative ability of the model. This close proximity to the top-left corner of the plot suggests that for a wide range of classification thresholds, we achieve

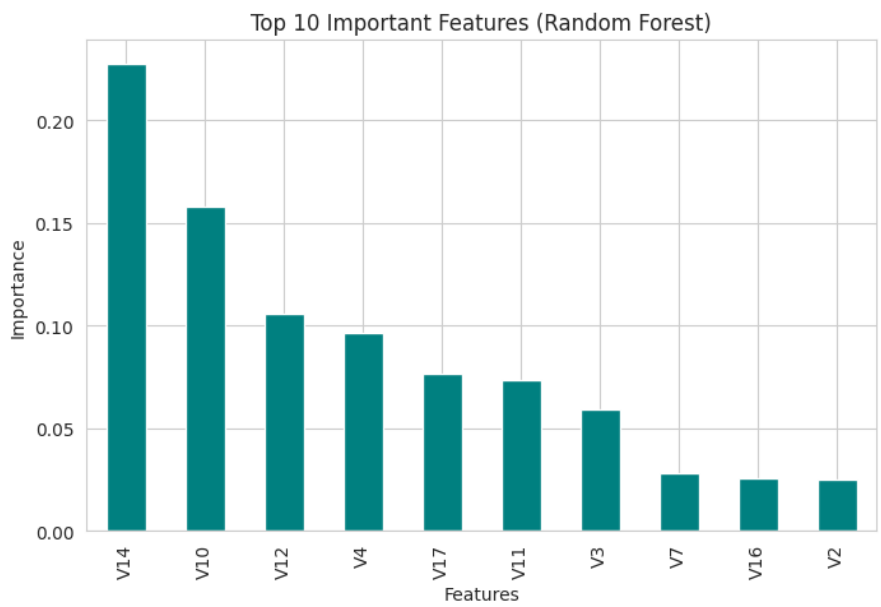
high true positive rates (TPR) with very low false positive rates (FPR). So the area under curve (AUC) is near to 1, which means classifier performance is almost perfect. The Random Forest model receives a higher sensitivity and specificity score than the Logistic Regression model in that it comes with a more desirable trade-off between performance metrics, making it better line tailored for this context.

Figure 19: Precision-Recall Curve-Random Forest



Precision-Recall curve gives us more insight into how well our model is doing in the presence of class imbalance. Despite increasing recall, Random Forest has very high precision which makes it suitable for finding rare fraudulent transactions without significantly increasing false positive. In imbalanced datasets specifically, traditional evaluation metrics like accuracy can be deceptive [8].

Figure 20: Feature Importance- Random Forest



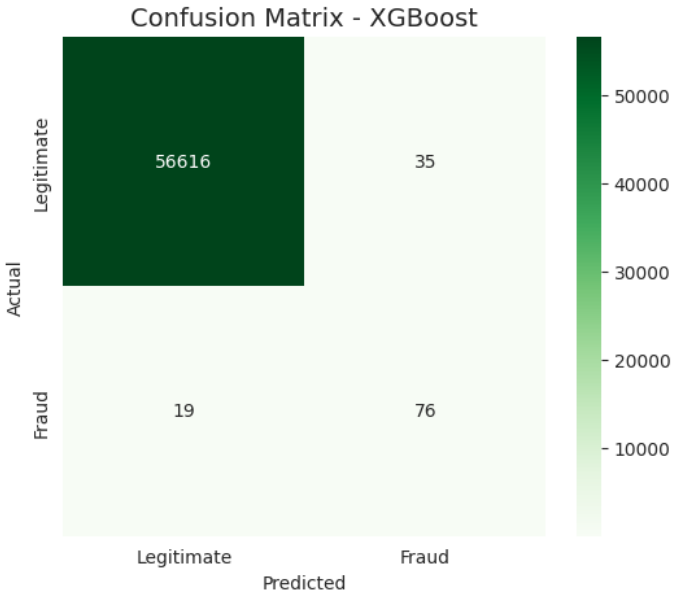
From the feature importance analysis, variables such as V14, V10 and V12 are contributing themost towards model prediction. The results here are consistent with those seen in exploratory data analysis phase, in which these features showed strong patterns for fraudulent transactions. To save space in this answer, I will have to skip the model output charts here, but there was a consistency between EDA and model output which makes you believe on your trained model more. It's a sign that the model is learning meaningful patterns rather than random noise.

In general, Random Forest greatly outscales performance of Logistic Regression as it deals with the imbalanced classes finely and encodes cross features very well. It is a simple and yet effective method, which can yield high precision while still preserving good re-call, making it a robust model for fraud detection." By exhibiting increased performance along with greater interpretability due to feature importance, Random Forest is one of the best performing models within this study and acts as a solid baseline for later improvements through more complex methods like boosting and hybrid modelling.

3.3.3 XGBoost

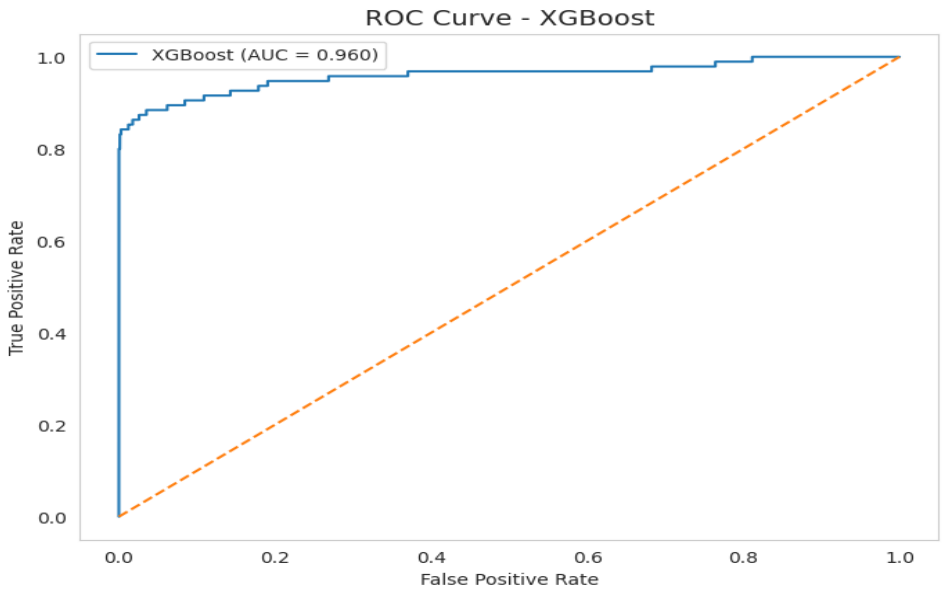
Finally, an advanced machine learning model, XGBoost (Extreme Gradient Boosting), was applied to improve the performance of credit fraud detection. In contrast to Random Forest, which constructs trees independently, XGBoost builds tree sequentially, where each new tree attempts to account for the errors made by previous trees. The boosting mechanism helps to improve performance by combining the predictions of a large number of weak learners, whereas regularization techniques help prevent overfitting [7]. XGBoost is one of the most popular algorithms for classification tasks, owing to its speed and capability to model complex interactions.

Figure 21: Confusion Matrix – XGBoost



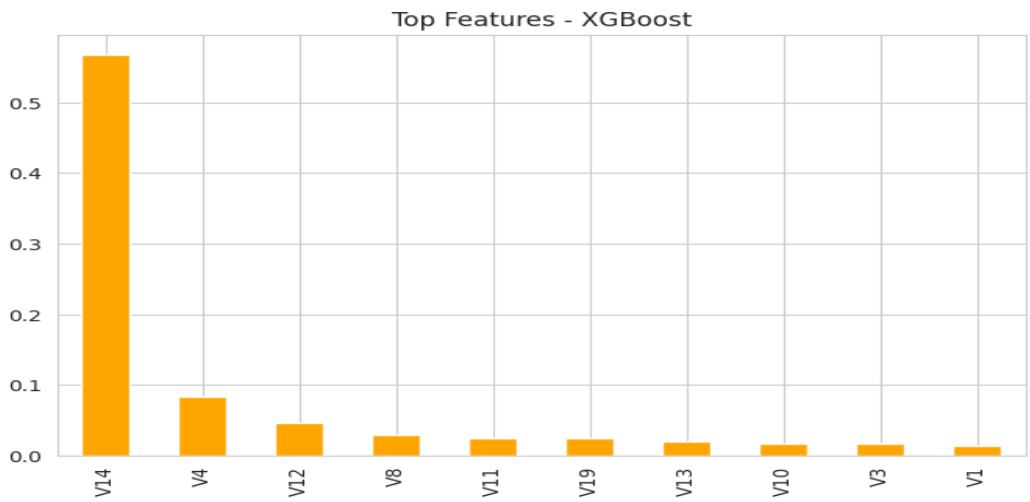
Confusion Matrix shows excellent classification performance The model correctly predicted 56,616 legitimate transactions with only 35 false positives, suggesting a high precision. Moreover, it detected 76 fraudulent transactions successfully and missed 19 cases. XGBoost performs better at detecting fraud (more true positives) compared to Random Forest while slightly increasing its false positive rate. This balance between precision and recall shows the effectiveness of the model in dealing with imbalanced data.

Figure 22 : ROC Curve – XGBoost



The ROC curve indicates a lovely performance, with an AUC score of about 0.96. Preferred curve stays near top-left corner, showing high true positive rate and low false positive rate at various thresholds. Although the Random Forest model achieved near-perfect performance, due to its boosting mechanism and regularization capabilities XGBoost provides overall a more stable and generalized model.

Figure 23: Feature Importance – XGBoost



Looking from the feature importance plot, we see that feature V14 constitutes the most dominant influence on model features followed by V4 & V12. This is in accordance with exploratory data analysis and results from Random Forest, which show these attributes are key to highlighting transactions that are fraud. The superior dominance of V14 also indicates the sensitivity of XGBoost towards most important Discriminative features to correctly separate fraudulent and legitimate patterns.

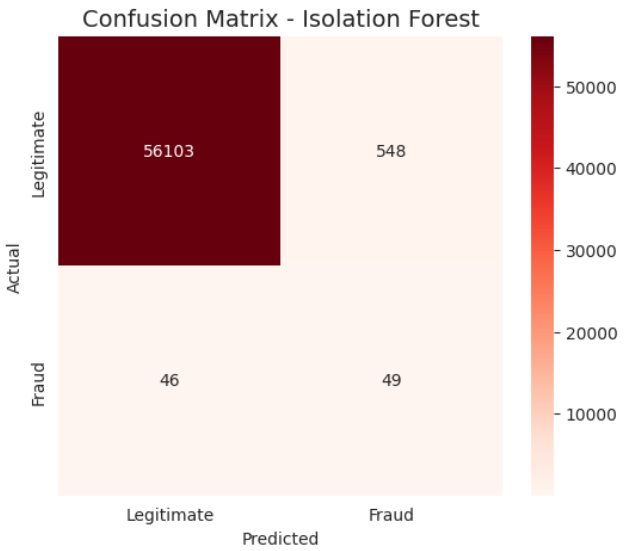
In short, XGBoost performs very competitively: it marginally outperforms Random Forest on detecting frauds while retaining a good generalization. The focus on hard-to-classify instances through boosting make it particularly useful in detecting rare occurrences of fraud. In addition, the robustness of important features across models strengthens the validity of findings.

Logistic Regression fared poorly with non-linear relationships and Random Forest improved performance by utilizing ensemble learning techniques. XGBoost enhances these benefits with boosting and regularization, refining it to an accurate model. This enables it as an essential part of a greater fraud detection system and makes the model an excellent choice for production systems.

3.3.4 Isolation Forest

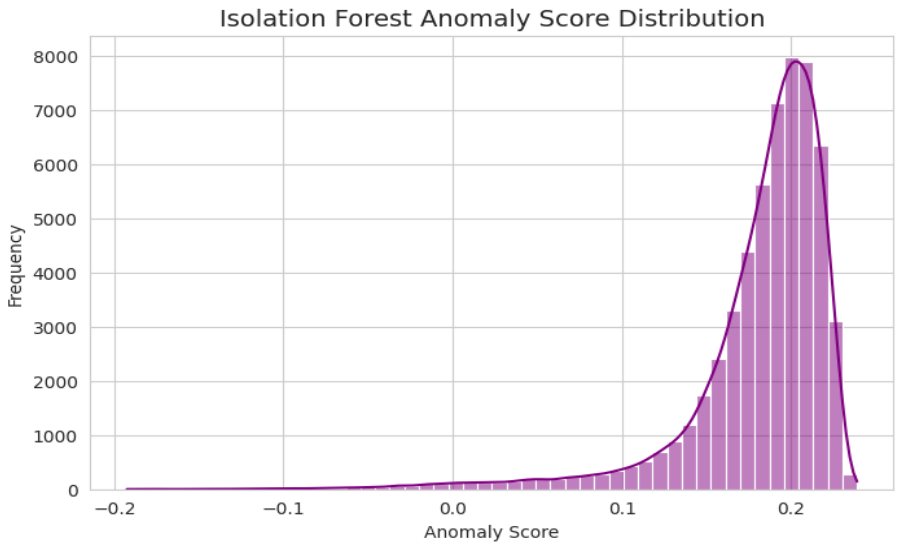
Isolation Forest is used as an Unsupervised anomaly detection method for fraudulent transactions detection. Isolation Forest does not require labeled data unlike supervised models like Logistic Regression, Random Forest and XGBoost. Instead, it identifies anomalies by treating observations that deviate significantly from the rest of the dataset as outliers. This method is rather appropriate for fraud detection as fraudulent transactions are infrequent and they have a differing behavior from normal ones [10].

Figure 24: Confusion Matrix – Isolation Forest



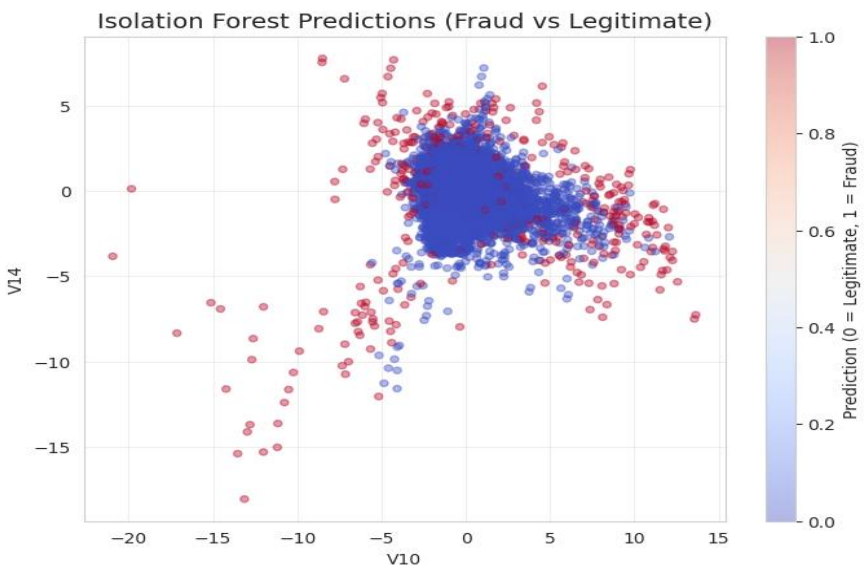
From the confusion matrix, we can see that it has 56,103 true positives in terms of legitimate transactions. But it did generate 548 false positives — much more than older models. The model detected 49 transactions as fraudulent while being unable to detect fraud in 46 cases. This indicates that Isolation Forest is able to detect anomalies, however the performance is less than that of supervised models because there are no label-based learning mechanisms available.

Figure 25: Anomaly Score Distribution – Isolation Forest



Intuition about separating normal and anomalous transactions can be derived from how the anomalous score distribution looks. The majority of transaction anomaly scores are clustered densely around a central range, where lower scores correspond to proposed anomalies. Fraudulent transactions are likely to show up in the tail of the distribution confirming that they fall out of normal behaviour. This distribution shows how well the model is able to assign appropriate anomaly scores for each transaction.

Figure 26 -Isolation Forest Predictions (Fraud vs Legitimate)



Scatter plot helps us see the difference of points between login and fraudulent transaction based on our selected features. Below: Legitimate transactions form dense clusters, while fraudulent individual transactions appear more scattered and isolated. This assumption aligns with the key wildlife of Isolation Forest — that outliers are few and unique. However, this overlap between classes necessitates the use of advanced techniques such as Mechanical Signals instead of just relying on anomaly detection to classify data points correctly.

In general Isolation Forest gives an effective idea about the nature of outliers in Reference : Although its single model classification performance lags behind that of supervised models, the ability to provide an anomaly score makes it very useful as a complementary method. The model well learns odd behavior and outliers, which are the defining traits of fraudulent transactions.

Logistic Regression and Random Forest are not only labeled models, but XGBoost uses boosting techniques. Isolation Forest takes another approach from anomaly detections perspective. Despite generating more false positives and missing some fraud cases, its outputs can be used for improvements to supervised models.

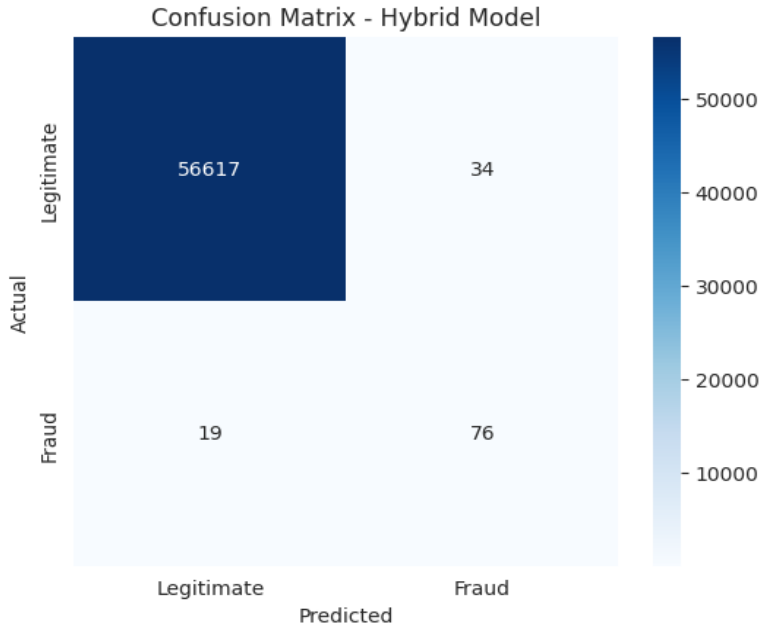
Isolated forest performs better when used along with the models instead of using it as a base classifier. This drives the establishment of a hybrid approach, where anomaly scores produced by Isolation Forest are integrated with more features in supervised machine learning models. The next section demonstrates the synergy of this combination to further boost fraud detection performance.

3.3.5 Hybrid Model (Isolation Forest + XGBoost)

To improve fraud detection, a hybrid model was developed that captured the advantages of both supervised and unsupervised learning approaches. Though XGBoost is designed to capture complex patterns from labeled data, Isolation Forests are responsible for detecting outliers without label dependencies. Simultaneously, the addition of anomaly scores produced by Isolation Forest as a feature in XGBoost serves to improve the model's ability in recognizing subtle and rare instances of fraud, thus resulting in efficient detection of these types of fraudulent transactions.

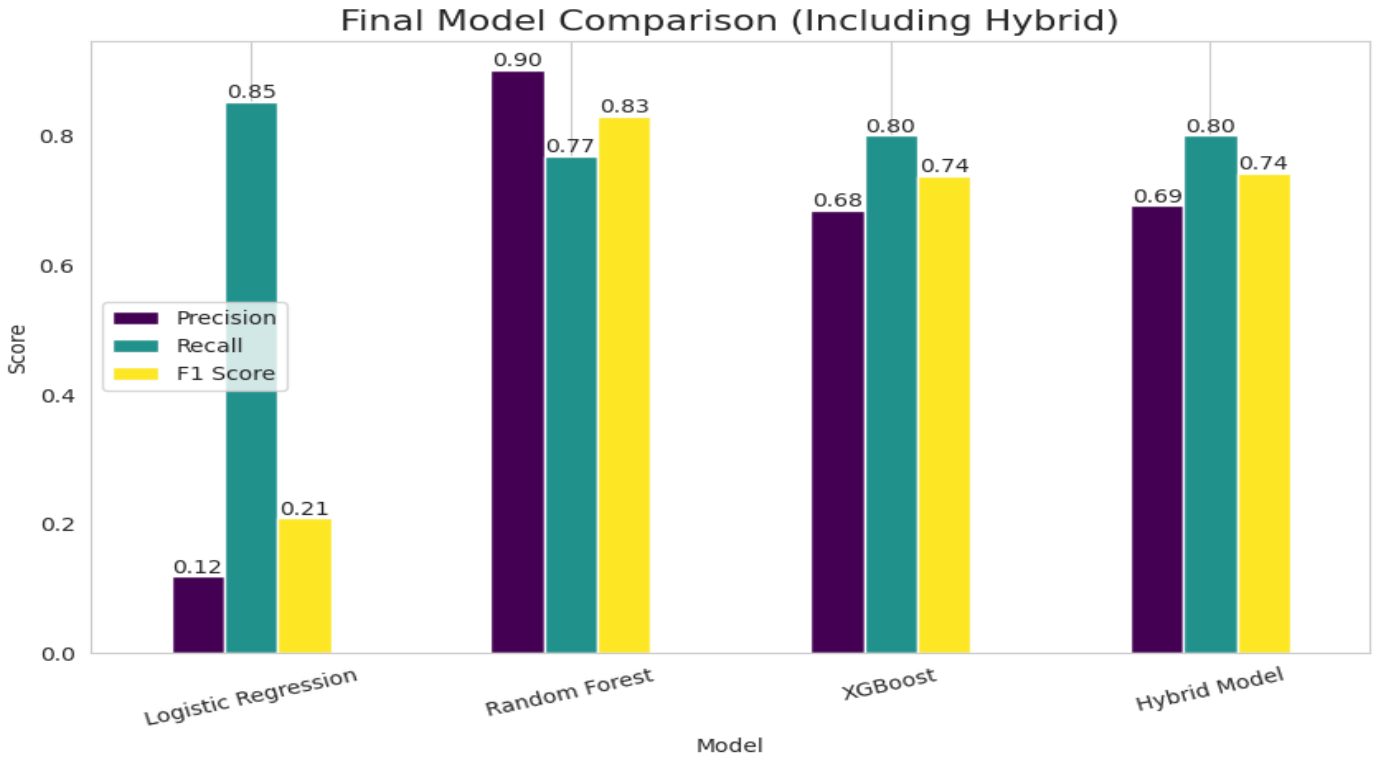
This amalgamation enables the model to harness both pattern recognition as well as outlier detection at once, thereby enhancing classification performance. The anomaly score serves as an extra signal that marks irregular transactions, which helps the model differentiate between normal and suspicious behaviour more effectively.

Figure 27: Confusion Matrix — Hybrid Model



We can see in the confusion matrix that our hybrid model classified 56617 legitimate transactions correctly with only 34 False Positives, indicating a strong precision. In addition, the model recognised 76 fraudulent transactions with only 19 false classifications. The hybrid approach provides an enhanced trade-off between precision and recall as compared with the state-of-the-art models, thus ensuring a better overall performance.

Figure 28: Comparison of Machine Learning Models(Precision, Recall, F1-Score)



The Model comparison chart depicts a complete assessment of all applied machine learning models with respect to precision, recall and F1-score. Due to the highly imbalanced nature of the dataset, accuracy is not an effective metric for performance evaluation, thus why these metrics are particularly important in fraud detection.

Logistic Regression which is the baseline model has a recall of 0.85, meaning it can detect a large number of fraudulent transactions. But its precision is very low (0.12), meaning that it wrongfully labels many real transactions as fraudulent. This brings about a low F1-score (0.21), demonstrating its incapacity to manage imbalanced data.

Similarity Matrix then passed into a Random Forest, which performs much better (the best precision is 0.90 and F1-score = 0.83) That means that it's very good at discarding false positives and still doing it with a strong fraud detection capability. Its even performance contributes to its reliability as one of the best models in this study.

XGBoost on the other hand provides a more balanced trade-off between precision (0.68) and recall (0.80) getting us an F1-score of 0.74. Although it is not as precise as Random Forest, we have seen that XGBoost has a great feature of generalization and stability caused by the boosting mechanism and regularization techniques.

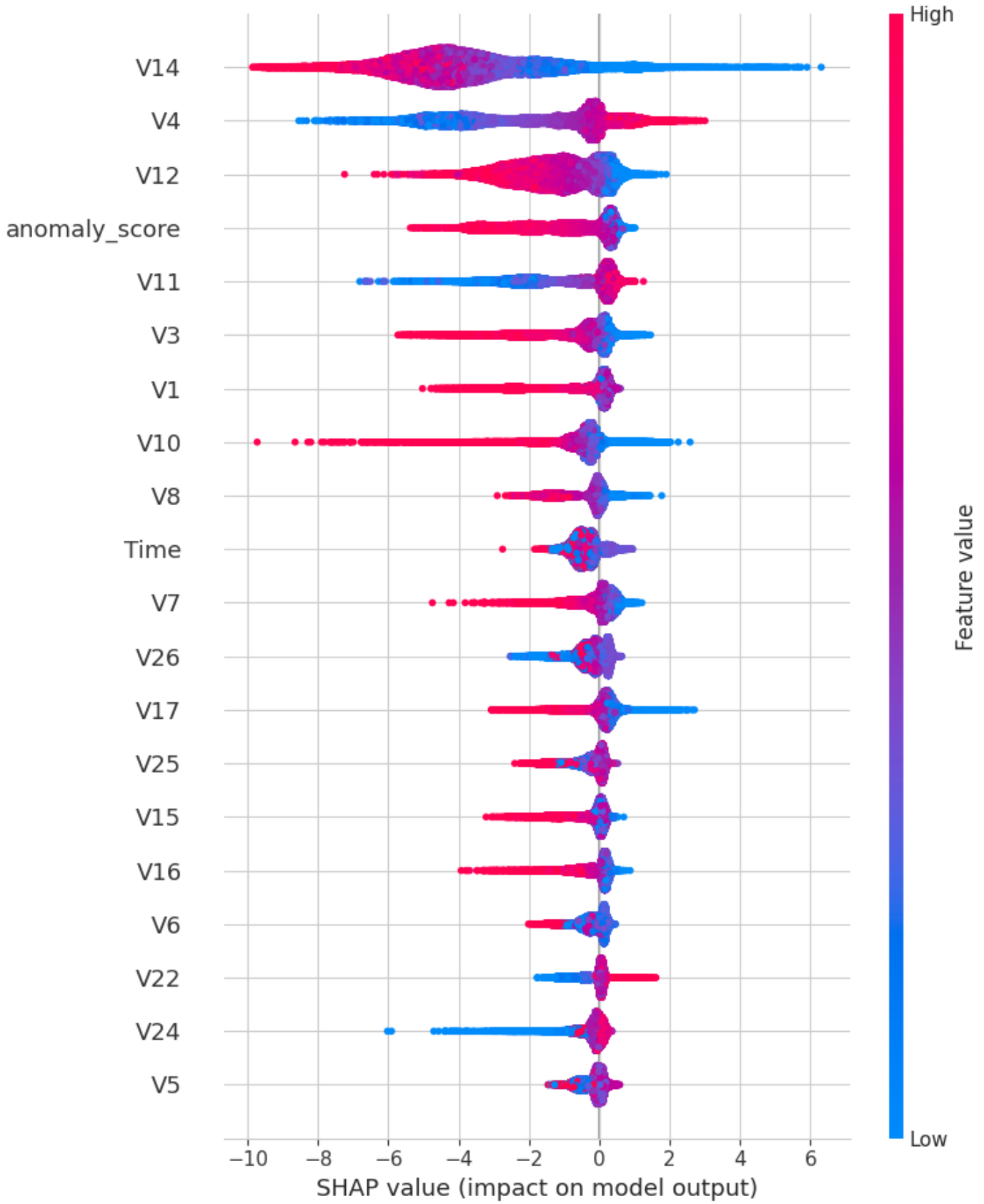
The hybrid model, which combines anomaly detection and supervised learning, resulted in a precision of 0.69, recall of 0.80 and an F1-score of 0.74 While it performed slightly worse than Random Forest in precision, it provides a much more stable balance on all metrics. Such a balance is particularly relevant in real-world fraud detection systems, where both the minimization of false positives and the maximization of up-voting fraudulent instances matters.

On the whole, this comparison shows that although Random Forest has a higher precision score than individual algorithms, the hybrid model outperforms on metrics such as sensitivity and specificity with more balanced results. This illustrates the usefulness of combining multiple methods to tackle problems with imbalanced and complex data sets!

3.3.7 Model Explainability using SHAP

SHAP (SHapley Additive exPlanations) was used to explain the final model predictions in order to improve model interpretability. It is a game-theoretic method that gives each feature importance values based on how well they contribute towards the model output. This enables a detailed understanding of the triggering features behind fraud detection decisions [11].

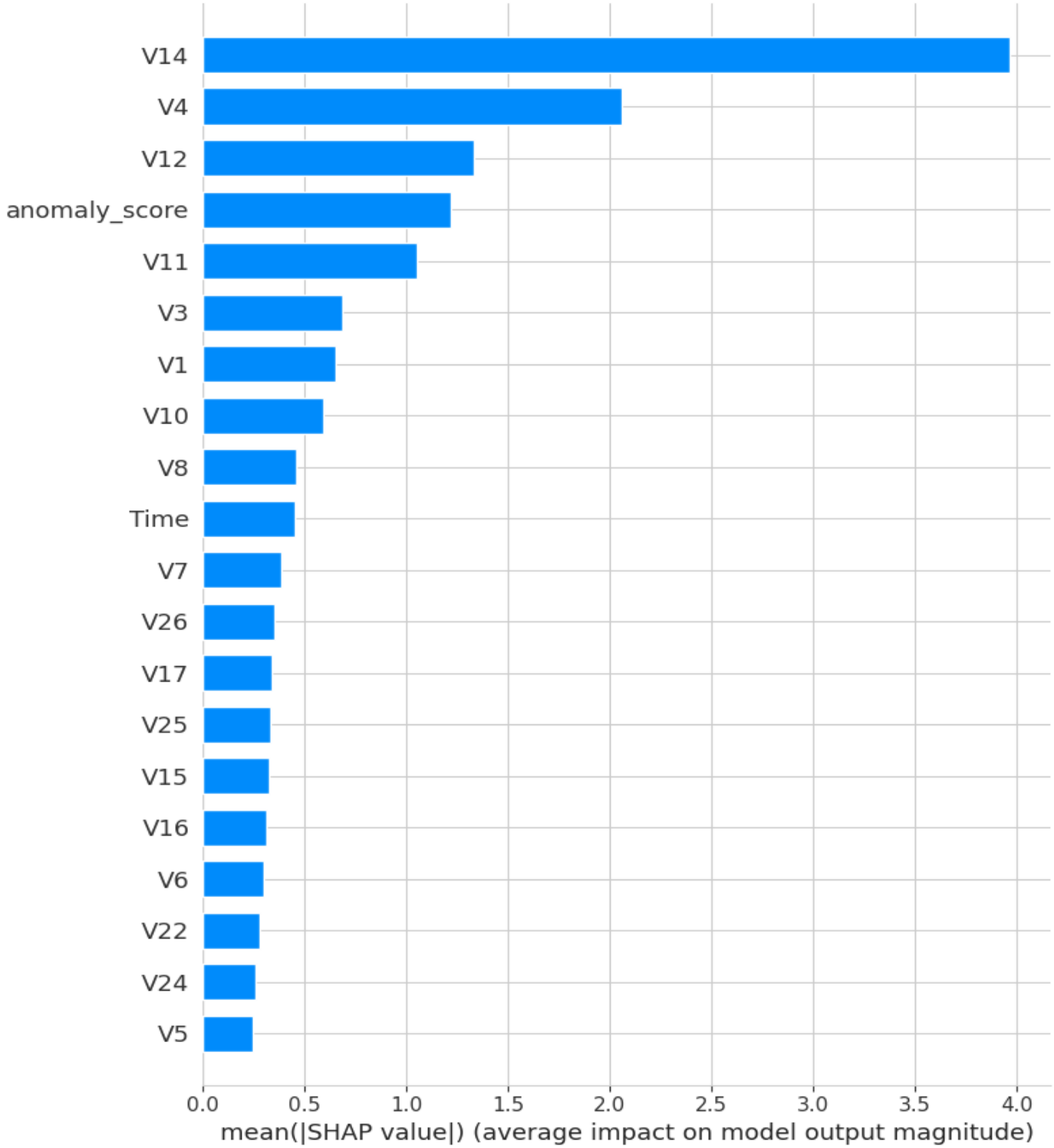
Figure 29- SHAP Summary Plot



The SHAP summary plot gives a global insight into features importance and their influence on model predictions. The color represents the feature value (low to high), and the position whether it impacts positively or negatively on the prediction (x axis) — each point represents an individual transaction.

V14, V4, and V12 features are the most powerful predictors of our model. Higher and lower feature values contribute differently to classification of fraud: we see non-linear relationships between features in the dataset.

Figure 30 - SHAP Feature Importance Plot



The bar chart of feature importance shows more clearly which were the most influential variables in the model. Importance Plot8 Confirm V14: Most Impact; Followed by 4 and 12 Confirm Random

Forest and XGBoost Result Also, that anomaly score generated from Isolation Forest was found to be one of the important features indicating the hybrid approach has indeed improved prediction capability.

SHAP gives useful information about how the model works internally, contributing to transparency and trust for the explainability field in general. One major advantage of SHAP is that it explains not just feature importance, but also direction (positive or negative), which makes it very suitable in critical applications like fraud detection (you want to know whether a more important feature contributes positively or negatively).

It's interesting to see that the key features were consistent across EDA, Random Forest, XGBoost and SHAP analysis which increases the reliability of the model. Moreover, the overall importance of anomaly scores among wizard feature scores affirms that combining both anomaly detection and supervised learning is effective.

SHAP provides both global and local interpretability, making the model's decisions understandable and justifiable. In financial systems, specifically, this explainability goes a long way for regulatory compliance as well as customer trust. Thus, SHAP is not mere a performance enhancements for the proposed system but also its prerequisite to have both interpretable and reliable environment for fraud detection.

Chapter 4: Evaluation and Results

4.1 Evaluation Metrics

The performance of the developed machine learning models was assessed using several evaluation metrics, including precision, recall, F1-score for classes and accuracy. And since the proportion of fraudulent transactions in our dataset is so small, accuracy will not be a good measure of performance.

Precision evaluates how many fraud transactions were correctly identified among all the predicted cases, and recall assess how well you could get a true positive fraudulent transaction. The F1-score combines precision and recall and is especially suited to imbalanced datasets [8]. Thus, these metrics ensure complete evaluation of a model performance in detecting fraud with lesser false alarms.

4.2 Model Performance Analysis

Classification reports, which evaluate the performance of each model in terms of precision, recall, and F1-score for both legitimate and fraudulent classes.

4.2.1 Logistic Regression Performance

Table 1: Logistic Regression Performance

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	1.00	0.99	0.99	56651
Fraud (1)	0.12	0.85	0.21	95
Accuracy			**0.99**	56746
Macro Avg	0.56	0.92	0.60	56746
Weighted Avg	1.00	0.99	0.99	56746

As shown in the output table below, our baseline model Logistic Regression shows a high recall of 0.85 for the fraud class means that our model detects most of the fraudulent transactions (also seen in further figure). Its precision, however, is very low (0.12), so many legitimate transactions are falsely identified to be fraudulent.] This leads to a very low F1-score of 0.21, indicating an imbalance between precision and recall. Since this is a common behaviour among imbalanced datasets (both for class or dataset)− models find it beneficial to overpredict and make few correct predictions on the minority class for maximizing recall. While Logistic Regression is effective for fraud detection, its high false positive rate renders it not amenable to real world applications where customer inconvenience must be minimized.

4.2.2 Random Forest Performance

The performance significantly improves on all metrics with Random Forest. It gets very high precision of 0.90 and recall of 0.77 for fraud transactions resulting in F1-score of 0.83. It shows that the model is very capable of avoiding false positives with being an effective fraud detecor. It

combines this high overall accuracy in part from its near-perfect classification of legitimate transactions.

Table 2: Random Forest Performance

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	1.00	1.00	1.00	56651
Fraud (1)	0.90	0.77	0.83	95
Accuracy			**1.00**	56746
Macro Avg	0.95	0.88	0.91	56746
Weighted Avg	1.00	1.00	1.00	56746

Finally, due to the ensembleing of data Random forest able to capture complex non-linear relationships with in data making it more robust over a baseline model. It is one of the most reliable models in this study, characterized by its capacity to balance between precision and recall.

4.2.3 XGBoost Performance

Table 3: XGBoost Performance

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	1.00	1.00	1.00	56651
Fraud (1)	0.68	0.80	0.74	95
Accuracy			**1.00**	56746
Macro Avg	0.84	0.90	0.87	56746
Weighted Avg	1.00	1.00	1.00	56746

XGBoost, in contrast, delivers a different, yet equally useful performance profile. This approach yields F1-score of 0.74: precision=0.68 and higher recall=0.80 In comparison with Random Forest, XGBoost trades off precision to give better recall, it catches more fraudulent transactions but also

has slightly higher false positives. In many systems, such as fraud detection, this trade off is desirable because the cost of missing an fraudulent transaction is greater than raising a false alarm. XGBoost achieves better recall performance as it iteratively boosts the classifier on hard-to-classify instances.

4.2.4 Isolation Forest Performance

Table 4: Isolation Forest Performance

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	1.00	0.99	0.99	56651
Fraud (1)	0.08	0.52	0.14	95
Accuracy			**0.99**	56746
Macro Avg	0.54	0.75	0.57	56746
Weighted Avg	1.00	0.99	0.99	56746

Compared with supervised anomaly detection methods, the performance of Isolation Forest, as an unsupervised model is noticeably poor. This results in precision of 0.08 and recall of 0.52 leading to a low F1-score (0.14). This model can only spot anomalies in the transaction without labeled data failing to identify whether its demands are fraudulent or legitimate. There are too many false positives indicating that a lot of normal transactions are being marked as anomalous. Despite this, the model is useful as it detects anomalies in the data.

4.2.5 Hybrid Model Performance

Table 5: Hybrid Model Performance

Class	Precision	Recall	F1-Score	Support
Legitimate (0)	1.00	1.00	1.00	56651
Fraud (1)	0.69	0.80	0.74	95
Accuracy			**1.00**	56746
Macro Avg	0.85	0.90	0.87	56746
Weighted Avg	1.00	1.00	1.00	56746

The hybrid model, combining the anomaly scores of Isolation Forest into XGBoost, yields a precision of 0.69 and recall of 0.80 with an F1-score of 0.74 This performance is similar to what we get with XGBoost but with added anomaly-based information. This hybrid method levels up from both supervised learning and anomaly detection to build a more robust model. Even though it has less accuracy than Random Forest, it gives a stable and balanced performance, so is very appropriate for use in practice applications.

4.3 Comparative Analysis

Table 6: Model Comparison Summary (Fraud Class Focus)

Model	Precision	Recall	F1-Score
Logistic Regression	0.12	0.85	0.21
Random Forest	0.90	0.77	0.83
XGBoost	0.68	0.80	0.74
Isolation Forest	0.08	0.52	0.14
Hybrid Model	0.69	0.80	0.74

By comparing the performance of all models, we can see significant variances in performance. This happens because Logistic Regression cannot deal with class imbalance effectively. Random Forest obtains the best precision and F1-score, hence being the most trustful model in terms of minimizing false positives.

Conclusion: XGBoost gives the best balance between precision and recall and thus better detection of fraudulent transactions. This balance is further refined in a hybrid model that builds on traditional anomaly detection approaches, further illustrating the advantages that can come from combining these different techniques.

Although Isolation Forest fails to predict the correct class as a standalone model, it provides relevant anomaly-based information, resulting in an increase of performance for the hybrid model.

4.4 Discussion of Results

The models also were able to detect the fraudulent transactions uniquely as opposed to traditional models, where one model detects all kinds of data. The Random Model and XG Boost potential datasets capture nonlinear relationships in observations/channel while the hybrid combined it with anomaly performance.

Data preprocessing was performed to remove duplicate transactions, normalize the data set, and partition it into training and test subsets. High precision minimizes false positives, while high recall ensures that fraud does not go unnoticed. The hybrid model strikes this balance well, which makes it a promising candidate for deployment in practice.

In general, all the results confirm that an ensemble of different machine-learning techniques yields a better and more robust solution for fraud detection.

Chapter 5: Conclusion

The objective of the project was to design a competent and efficient machine learning system for detecting fraudulent financial transactions. Considering the class imbalance problems within the dataset, this study explored several ways, including traditional models, ensemble methods (FLD + ELM), anomaly detection algorithms as well as hybrid through a combination of supervised and unsupervised learning.

From the results, it has been unfolded that when appropriate strategies are employed, machine learning techniques become very optimized in fraud detection. Logistic Regression: While still providing a good baseline, it struggles to balance the precision & recall with class imbalance. It was able to return many relevant links but with extremely low precision, thus not deployable in a real world applications.

Ensemble methods like Random Forest and XGBoost helped improve performance a lot. As it had the most highly dependable outcomes with respect to fewer false positives, Random Forest developed the higher precision and F1 rating. On the other hand, XGBoost also maintained a clear trade-off but gave a reasonable balance between precision and recall much better than antenna each torch model Overall strongest ability to identify fraudulent transaction

Isolation Forest was incorporated into the system, enabling the detection of anomalous transaction patterns. While not performing particularly well on its own, the inclusion of Isolation Forest in a hybrid model improved the overall robustness that helped boost system performance.

The hybrid model was gently positioned as an ideal compromise, capturing strengths of both anomaly detection and supervised learning approaches. It obtained a good trade-off between precision and recall, which well-suited for the real-world fraud detection systems in-demand, where false positives and missed fraud lead to large losses.

Also, our implementation of SHAP for model interpretability gave us valuable information as it is related to which features are more important than others ensuring transparency and trust in the predictions made by the model. Again, V14, V4 and V12 appeared to be the most important features in classifying a transaction as fraudulent.

In summary, the project clearly illustrates how the merging of many machine learning approaches with careful management of class imbalance and interpretability creates an accurate and professional level fraud detection framework.

5.1 Future Work

Although the proposed system provided strong performance, there are several elements of improvement that we can explore in future work to enhance its effectiveness and applicability.

For example, you can implement better models in a transactional way like deep learning based approaches to learn complex patterns from your dataset (Neural networks or even pass the data through an autoencoder). In particularly dynamic settings, they could also improve fraud detection accuracy.

An essential direction is the real-time fraud detection. We trained the model using batch processing, in a production system once deployed we can deploy this model to catch fraudulent transactions instantaneously.

Other techniques for handling imbalanced data, such as adaptive resampling strategies or generative models to account for the imbalance, can also be explored to enhance application performance. Hyperparameter tuning and automated machine learning (AutoML) techniques can help to more efficiently optimize model performance as well.

For instance, including new input methods or contextual information could complement the current approach by adding to signal extraction in behavioral detection time series. Besides, connecting the system with financial platforms and evaluating it in real life will boost its practicality.

Lastly, enhanced interpretability techniques can improve explainability and build trust amongst stakeholders and or fulfill compliance with regulatory obligation.

5.2 Reflection

There were a few important lessons to learn regarding the difficulties and nuances of using machine learning for fraud detection from this project. This – along with handling imbalanced data, where accuracy will not be as reliable of an evaluation metric.

By deploying multiple models, we could gain more insight on their strengths and weaknesses. We saw the limitations of simple models with Logistic Regression, and then the power of combining multiple decision trees with ensemble methods. In addition, the deployment of XGBoost highlighted the model efficiency issue by leveraging the boosting technique during performance improvement.

One of the main outcomes of this project being to build a hybrid model that combined anomaly detection with supervised learning. Not only did this approach improve performance, but it also exhibited the advantage of using multiple methodologies in conjunction with one another.

The last but an important thing to do in this project was SHAP, model interpretability. Interpreting how features contribute to predictions is very significant in financial applications, and this collaborative experience reinforced the value of transparency in machine learning systems.

On a practical note, the project improved skills in data preprocessing and model development/evaluation/visualization. It also proved the value of critical thinking in choosing appropriate models and interpreting results.

In summary, this project has offered me a comprehensive understanding of using machine learning for fraud detection as well as the need for bridging those use cases with more practical aspects of developing these solutions in real-world scenarios.

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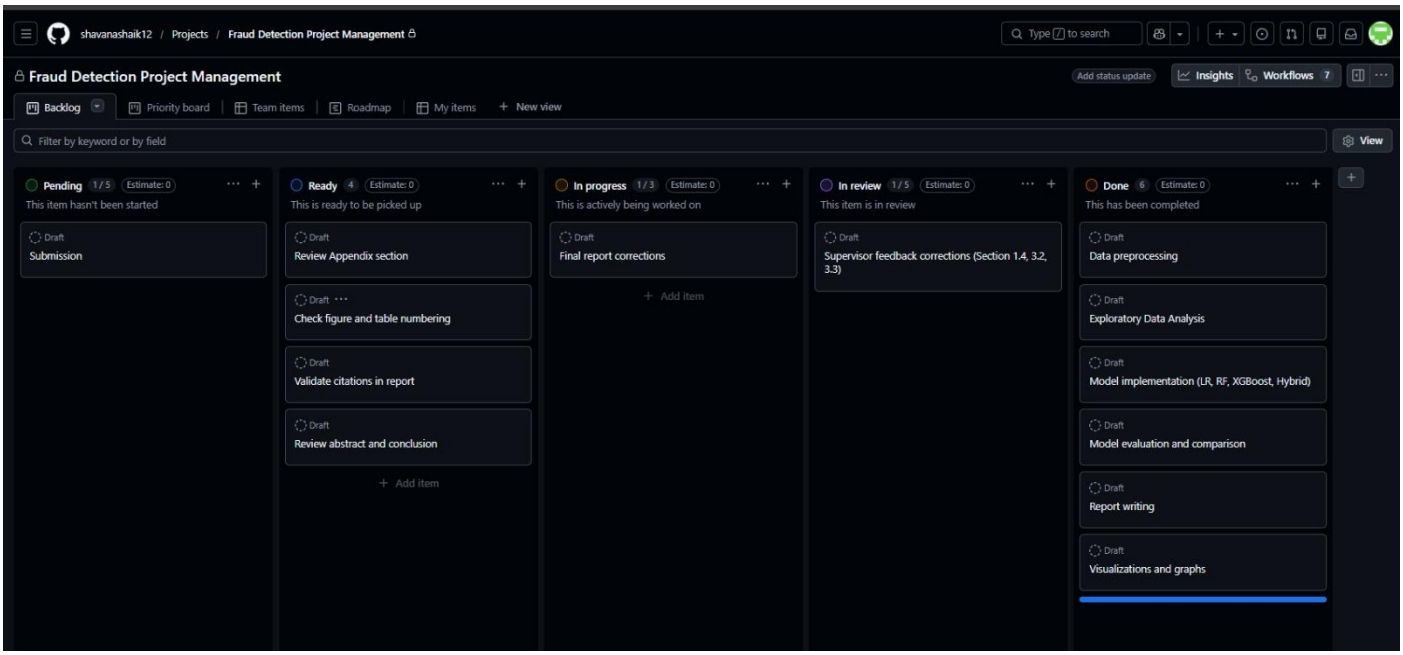
7. Appendices

Appendix A : Kanban Board for Project Management

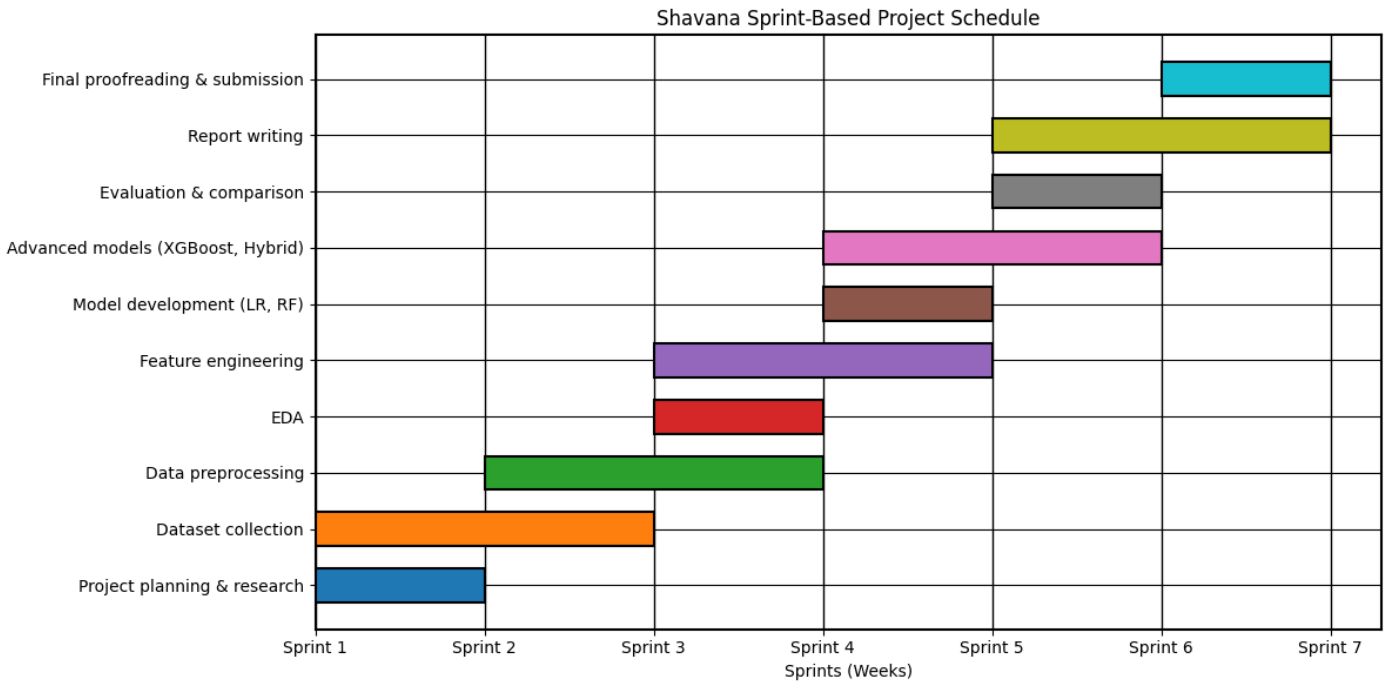


In this appendix, we show the Kaban board in our GitHub Projects to manage and track the tasks around the project. The Workflow was divided into five stages: Pending, Ready, In Progress, In Review and Done.

Tasks were sorted into columns based on status and moved from one to another as the project progressed. The tasks that were completed, such as data preprocessing and exploratory data analysis (EDA), model implementation, and evaluation of the models were marked under Done. You place any task which requires to be validated, including correction of veteran feedback and report improvements again into the In Review stage. The In Progress column was for the tasks already underway including final report corrections while pending and ready were reserved for remaining issues such as preparation steps prior to submission.

Using this kanban-based approach, we had a good visibility of where tasks Click to copy task deployment Ubuntu into the project timeline. It also complemented the sprint-based schedule by providing a day-to-day, task-level progress view of how things were going in the final stages of projects.

Appendix B: Sprint-Based Project Schedule



Appendix B: This appendix shows the project schedule that is organised using sprints for tracking the general progress of the project. The project was split into a number of sprints, and each sprint is for certain period of time completed with certain tasks.

The first few sprints comprised of project planning, research projects and datasets acquisition as well as data processing. These stages formed the basis for further analysis. Next, the middle sprints were focused on exploratory data analysis and feature engineering -> then fitted Machine Learning Models like Logistic Regression, Random Forest.

Advanced models like XGBoost and hybrid approaches were adopted in all sprints after that and performance was evaluated. Performance analysis, report writing and final proofing/submission process were earmarked for the end of the project. By utilizing a sprint-based schedule, I was able to create an organized workflow that effectively managed time and accomplished all project milestones in a systematic manner.

Appendix C: **Please Click here for Code Link Via GitHUB**

Appendix D: **Please click here for Dataset Link**

Appendix E: **Please click here for the youtube link of Python code or Artefact by video explianation**